

A Radial–Angular Laplace Principle for Homogeneous Functionals

Enkelejd Hashorva

Abstract

We study joint upper tails of a fixed finite family of positively homogeneous functionals of a random vector with independent radial and angular components. Conditioning on the angle reduces the event to a single radial exceedance whose required radius is the maximum of the component costs. For radial laws in the Gumbel max-domain of attraction, verified local normal forms yield exact tail equivalents and conditional angular and excess laws. Smooth minimizing manifolds are allowed. Slack constraints, weak contacts, and genuinely switching constraints produce Gaussian, truncated Gaussian, and mixed linear–quadratic limits, respectively. A weighted formulation covers moving thresholds and singular angular densities under explicit local convergence and global tightness conditions. We recover the scalar Gaussian-chaos equivalent, verify a Dirichlet specialization, and give explicit finite-grid Gaussian and Brownian examples. A separate chi-square benchmark yields a uniform tail approximation and the conditional excess scale when the input dimension grows with the threshold. The angular results remain first-order transfer theorems under supplied normal forms.

Keywords: Laplace method; homogeneous functionals; Gumbel domain of attraction; Gaussian chaos; minimizing manifold; conditional excesses.

1 Introduction

For a smooth scalar homogeneous functional, radial–angular tail asymptotics are governed by the minimum of one smooth required-radius function on the angular space. A vector upper-tail event has a different local geometry. Even if every component functional is smooth, conditioning on the angle produces the maximum of finitely many required-radius functions. This maximum is generally nonsmooth at precisely those angles where several output constraints are relevant.

The two-output case is the smallest setting in which the distinction is visible. It contains three basic mechanisms. If one constraint is strictly slack, the usual quadratic Laplace approximation applies. If the second constraint ties but has zero Karush–Kuhn–Tucker multiplier, it truncates the Gaussian normal tube without changing the power. If both multipliers are positive, one normal direction has first-order contact and is scaled differently from the remaining quadratic directions. For $d > 2$, several half-spaces can form a Gaussian cone, or a switching space of rank greater than one can acquire the faster linear scale. These descriptions require the nondegeneracy assumptions stated below; active-set labels alone do not determine a local normal form.

The smooth scalar case is classical in this setting. Korshunov, Piterbarg and Hashorva [7, Theorem 3] derive Laplace expansions near a smooth minimum manifold and obtain Weibullian and Gaussian chaos asymptotics in their Theorem 5 and Corollary 7. Hashorva, Korshunov and Piterbarg [8, Theorems 1.2 and 1.3] obtain Gaussian tail and density expansions and a first-order extension to a Gumbel radial law with a positive continuous angular density. Our scalar specialization recovers their leading term. The emphasis here is the maximum of finitely many required-radius branches, its different normal scales, and angular measures that may vanish or be singular on the minimizing set.

Ievlev [9] develops a broader geometric Laplace calculus using anisotropic normal cells near stratified minimum sets. His separation of local rescaling from control of the full integral is a useful comparison for our weighted formulation. We retain a smooth reference manifold and explicitly verify the local limits and tightness in the examples; incident strata, frontier rescalings, and higher-order expansions are outside the present scope.

Our purpose is to isolate these mechanisms in a first-order theorem and to test the theorem first on finite-dimensional nonnegative Gaussian quadratic forms and then on squared Brownian norms at two and at finitely many times. When all outputs have the same degree and share a common threshold, the angular and radial limits also determine the conditional excess vector on the corresponding common output scale, under branchwise phase convergence. In particular, they determine the asymptotic location of its minimum under almost-sure uniqueness of the limiting branch, and the minimum excess then separates as an independent exponential variable.

On a fixed grid, a centered Gaussian process is determined by its covariance matrix; Brownian motion is special only because its increment structure makes the relevant cones explicit. We state a covariance-only anchor condition under which an arbitrary finite-grid chi-square process has the same weak-cone structure. The examples are stress tests: they show that the number of active constraints alone does not determine the power, and that rank loss or coalescing constraints fall outside a fixed normal form. Density approximations and higher-order expansions for general functionals are deliberately excluded.

Section 6 adds an elementary quantitative benchmark for the same quadratic rule in growing input dimension. It records both a uniform relative error and the resulting excess scale. The smooth high-dimensional Laplace results of [10, 11] provide the relevant broader comparison; no dimension-uniform angular theorem is asserted. The main proofs are collected in Section 7, while the short benchmark is proved with its statement. Appendices A–C contain the weighted verification criteria, the detailed Dirichlet specialization, and the two-time Brownian calculations.

2 Exact radial reduction

Let $\mathcal{S} \subset \mathbb{R}^k$ be a compact $n_{\mathcal{S}}$ -dimensional C^2 angular section: every ray of the cone under consideration meets $\mathcal{S}$ exactly once. Thus scalar multiplication is defined on $\{ru : r > 0, u \in \mathcal{S}\}$. Let U have probability law μ on $\mathcal{S}$. Let $R > 0$ be independent of U , with distribution function F , infinite upper endpoint, and survival function $\bar{F}(x) = \mathbb{P}\{R > x\}$. Suppose that, for $j = 1, \dots, d$,

$$G_j(ru) = r^{\gamma_j} g_j(u), \quad r > 0, \quad u \in \mathcal{S}, \quad (2.1)$$

where $\gamma_j > 0$ and g_j is continuous. Fix $\mathbf{b} = (b_1, \dots, b_d) \in (0, \infty)^d$ and consider the degree-matched thresholds

$$E_{\tau}(\mathbf{b}) = \bigcap_{j=1}^d \{G_j(RU) > b_j \tau^{\gamma_j}\}, \quad \tau \rightarrow \infty. \quad (2.2)$$

For $u \in \mathcal{S}$ define each branch separately by

$$q_j(u) = \begin{cases} (b_j/g_j(u))^{1/\gamma_j}, & g_j(u) > 0, \\ \infty, & g_j(u) \leq 0, \end{cases} \quad q(u) = \max_{1 \leq j \leq d} q_j(u). \quad (2.3)$$

We use the convention $\bar{F}(\infty) = 0$. Assume that the common positivity set $\bigcap_{j=1}^d \{g_j > 0\}$ is nonempty. Compactness and lower semicontinuity of the extended costs then imply $0 < q_* := \min_{u \in \mathcal{S}} q(u) < \infty$. Conditioning on U yields the exact identity

$$\mathbb{P}\{E_{\tau}(\mathbf{b})\} = \int_{\mathcal{S}} \bar{F}(\tau q(u)) \mu(du). \quad (2.4)$$

Thus all output geometry enters through the scalar, event-dependent cost $q = \max_j q_j$. When $d = 2$, reparametrizing τ allows us to write $\mathbf{b} = (1, b)$; we then put $E_{\tau}(b) = E_{\tau}((1, b))$.

The same reduction remains exact for moving thresholds. Given an eventually positive vector $\mathbf{t}_\tau = (t_{\tau,1}, \dots, t_{\tau,d})$, put

$$\rho_{\tau,j}(u) = \begin{cases} (t_{\tau,j}/g_j(u))^{1/\gamma_j}, & g_j(u) > 0, \\ \infty, & g_j(u) \leq 0, \end{cases} \quad \rho_\tau(u) = \max_{1 \leq j \leq d} \rho_{\tau,j}(u), \quad (2.5)$$

and define

$$E_\tau(\mathbf{t}_\tau) = \bigcap_{j=1}^d \{G_j(RU) > t_{\tau,j}\}. \quad (2.6)$$

Then

$$\mathbb{P}\{E_\tau(\mathbf{t}_\tau)\} = \int_{\mathcal{S}} \bar{F}(\rho_\tau(u)) \mu(du). \quad (2.7)$$

The fixed-ray setting is recovered by $t_{\tau,j} = b_j \tau^{\gamma_j}$, in which case $\rho_\tau(u) = \tau q(u)$.

Write

$$\mathcal{M} = \{u \in \mathcal{S} : q(u) = q_*\}, \quad r_\tau = \tau q_*. \quad (2.8)$$

We assume that F belongs to the Gumbel max-domain of attraction with auxiliary function w , that is,

$$\frac{\bar{F}(r + x/w(r))}{\bar{F}(r)} \longrightarrow e^{-x}, \quad x \in \mathbb{R}, \quad r \rightarrow \infty, \quad (2.9)$$

see, for example, [2], and set

$$\lambda_\tau = r_\tau w(r_\tau). \quad (2.10)$$

For an infinite-endpoint Gumbel law, $1/w(r) = o(r)$ as $r \rightarrow \infty$; hence $\lambda_\tau \rightarrow \infty$ as $\tau \rightarrow \infty$.

The local arguments below use two standard localization conditions. First, for every sufficiently small neighborhood V of $\mathcal{M}$ there is $\eta_V > 0$ such that

$$\inf_{u \notin V} \frac{q(u)}{q_*} \geq 1 + \eta_V. \quad (2.11)$$

This separation is automatic here: the extended cost q is lower semicontinuous on the compact set $\mathcal{S}$, and $\mathcal{M}$ is precisely its minimizer set. We record it because its uniform analogue is a genuine condition for moving thresholds. Second, a standard one-sided bound for Gumbel tails states that, for every fixed $p > 0$, there is $C_p < \infty$ such that

$$\frac{\bar{F}(r_\tau(1 + x/\lambda_\tau))}{\bar{F}(r_\tau)} \leq C_p(1 + x)^{-p}, \quad x \geq 0, \quad (2.12)$$

for all sufficiently large τ . Here is a direct justification that also specifies the uniformity in x . The Gumbel representation [2] permits

$$\bar{F}(r) = c(r) \exp \left\{ - \int_{r_0}^r \frac{dt}{a(t)} \right\}, \quad c(r) \rightarrow c > 0, \quad a(r)w(r) \rightarrow 1,$$

as $r \rightarrow \infty$, where $a > 0$ is absolutely continuous and $a'(r) \rightarrow 0$ as $r \rightarrow \infty$. For any $\epsilon > 0$, eventually $a(t) \leq a(r) + \epsilon(t - r)$ for every $t \geq r$. Hence

$$\frac{\bar{F}(r + s)}{\bar{F}(r)} \leq C \left(1 + \frac{\epsilon s}{a(r)} \right)^{-1/\epsilon}, \quad s \geq 0.$$

Taking $s = x/w(r)$ and $1/\epsilon > p$ proves (2.12). This is consistent with the multiplicative rapid-variation estimates in [1]. Together with (2.11), it also controls the angular complement: there $x = \lambda_\tau(q/q_* - 1)$ is bounded below by a positive multiple of λ_τ . The bound holds, in particular, for the chi radial distributions used below. Recording it explicitly separates the local angular expansion from control of the full integral in (2.4).

3 A finite-output normal-form theorem

For the first theorem in this section, assume that $\mathcal{M}$ is a compact, boundaryless, m -dimensional C^2 submanifold of $\mathcal{S}$, and put $c = n_{\mathcal{S}} - m \geq 1$. Let $N_s\mathcal{M}$ denote the normal space of $\mathcal{M}$ inside $\mathcal{S}$ and define the total normal bundle

$$\mathbf{N}\mathcal{M} = \{(s, x) : s \in \mathcal{M}, x \in N_s\mathcal{M}\},$$

equipped with bundle volume $dV_{\mathcal{M}}(s) dx$. Let $\mathcal{T}$ be a fixed tubular neighborhood of $\mathcal{M}$, and let $\Psi(s, z)$ be its tubular chart with $z \in N_s\mathcal{M}$, chosen so that its volume Jacobian is one at $z = 0$. Suppose for this theorem that μ has a continuous positive density h near $\mathcal{M}$ with respect to the induced Riemannian volume on $\mathcal{S}$. Set

$$f_j = \frac{q_j}{q_*}, \quad f = \frac{q}{q_*} = \max_{1 \leq j \leq d} f_j.$$

The following formulation deliberately separates the Laplace argument from the classification of the nonsmooth minimum. For $\lambda \geq 1$, let $D_{\lambda,s} : N_s\mathcal{M} \rightarrow N_s\mathcal{M}$ be a continuous family of invertible linear maps such that

$$|\det D_{\lambda,s}| = \lambda^{-\kappa}, \quad \kappa > 0. \quad (3.1)$$

The maps may contract different normal directions at different rates. We also require every normal direction to contract uniformly:

$$\sup_{s \in \mathcal{M}} \|D_{\lambda,s}\| \rightarrow 0, \quad \lambda \rightarrow \infty. \quad (3.2)$$

Suppose that there are phases $\Phi_s : N_s\mathcal{M} \rightarrow [0, \infty]$ such that $(s, x) \mapsto \Phi_s(x)$ is Borel measurable and, for bundle-volume almost every (s, x) ,

$$\lambda [f(\Psi(s, D_{\lambda,s}x)) - 1] \xrightarrow{\lambda \rightarrow \infty} \Phi_s(x). \quad (3.3)$$

Here and below $e^{-\infty} = 0$. Expressions involving $\Psi(s, D_{\lambda,s}x)$ are restricted to the expanding domain on which the chart is defined; in view of (3.2), every compact subset of $\mathbf{N}\mathcal{M}$ is contained in that domain for all large λ . Assume that there are $c_0 > 0$ and jointly Borel measurable, uniformly coercive gauges $\varrho_s : N_s\mathcal{M} \rightarrow [0, \infty)$ satisfying

$$\lambda [f(\Psi(s, D_{\lambda,s}x)) - 1] \geq c_0 \varrho_s(x), \quad (3.4)$$

$$\sup_{s \in \mathcal{M}} \int_{N_s\mathcal{M}} (1 + \varrho_s(x))^{-p} dx < \infty, \quad (3.5)$$

whenever $D_{\lambda,s}x$ belongs to $\mathcal{T}$, for some $p > \kappa$. The preceding bounds imply that the following integral is finite; assume that it is positive:

$$C_{\Phi} := \int_{\mathcal{M}} h(s) \int_{N_s\mathcal{M}} e^{-\Phi_s(x)} dx dV_{\mathcal{M}}(s) > 0. \quad (3.6)$$

Theorem 3.1. *Under (2.9) and (3.1)–(3.6), with the order $p > \kappa$ specified above, we have*

$$\mathbb{P}\{E_{\tau}(\mathbf{b})\} \underset{\tau \rightarrow \infty}{\sim} \bar{F}(r_{\tau}) \lambda_{\tau}^{-\kappa} C_{\Phi}. \quad (3.7)$$

On the fixed tube, let $S_{\tau} = \pi(U)$ be the tubular projection and let z_{τ} be the normal coordinate of U ; extend both variables arbitrarily outside the tube. The conditional probability of that complement tends to zero. Conditionally on $E_{\tau}(\mathbf{b})$,

$$\left(S_{\tau}, D_{\lambda_{\tau}, S_{\tau}}^{-1} z_{\tau}\right) \xrightarrow{\tau \rightarrow \infty} \mathbf{Q}_{\Phi}, \quad \mathbf{Q}_{\Phi}(ds, dx) = C_{\Phi}^{-1} h(s) e^{-\Phi_s(x)} dV_{\mathcal{M}}(s) dx. \quad (3.8)$$

In fact, the angular convergence in (3.8) holds in total variation. Moreover,

$$w(r_\tau)(R - \tau q(U)) \mid E_\tau(\mathbf{b}) \xrightarrow[\tau \rightarrow \infty]{d} \text{Exp}(1), \quad (3.9)$$

independently of the limiting angular variables.

Corollary 3.2. Assume the conditions of Theorem 3.1 and $d \geq 2$. Suppose in addition that

$$\gamma_1 = \cdots = \gamma_d =: \gamma, \quad \mathbf{b} = \mathbf{1}_d,$$

and put

$$M_\tau := \min_{1 \leq j \leq d} G_j(RU), \quad J_\tau := \min_{1 \leq j \leq d} \text{argmin } G_j(RU).$$

Let $(S, X) \sim \mathbf{Q}_\Phi$, let E be a unit exponential random variable independent of (S, X) , and set

$$\zeta_\tau := \frac{q_* w(r_\tau)}{\gamma \tau^{\gamma-1}} = \frac{\lambda_\tau}{\gamma \tau^\gamma}. \quad (3.10)$$

Then, conditionally on $E_\tau(\mathbf{1}_d)$,

$$\zeta_\tau(M_\tau - \tau^\gamma) \xrightarrow[\tau \rightarrow \infty]{d} E. \quad (3.11)$$

To identify the full output vector and the location, define, for $\lambda \geq 1$ and on the expanding tubular domain, the vector of individual branch phases by

$$\varphi_\lambda(s, x) := (\lambda [f_j(\Psi(s, D_{\lambda,s}x)) - 1])_{j=1}^d. \quad (3.12)$$

Suppose that there is a Borel measurable map $\varphi : \mathbf{NM} \rightarrow [-\infty, \infty]^d$ such that

$$\varphi_\lambda(s, x) \xrightarrow{\lambda \rightarrow \infty} \varphi(s, x) \quad \text{for } \mathbf{Q}_\Phi\text{-almost every } (s, x), \quad (3.13)$$

in the product extended-real topology. Write its components as $\varphi_{j,s}(x)$. Since the maximum phase already converges,

$$\max_j \varphi_{j,s}(x) = \Phi_s(x) < \infty \quad \mathbf{Q}_\Phi\text{-almost surely.}$$

Then, conditionally on $E_\tau(\mathbf{1}_d)$,

$$(\zeta_\tau(G_j(RU) - \tau^\gamma))_{j=1}^d \xrightarrow[\tau \rightarrow \infty]{d} (E + \Phi_S(X) - \varphi_{j,S}(X))_{j=1}^d \quad (3.14)$$

in $[0, \infty]^d$, where subtracting $-\infty$ gives $+\infty$. If, in addition, the limiting maximum is attained at a unique index $\mathbf{Q}_\Phi$ -almost surely, define

$$J_* := \text{argmax}_{1 \leq j \leq d} \varphi_{j,S}(X).$$

Then

$$(\zeta_\tau(M_\tau - \tau^\gamma), J_\tau) \xrightarrow[\tau \rightarrow \infty]{d} (E, J_*), \quad E \text{ independent of } J_*. \quad (3.15)$$

In particular,

$$\mathbb{P}\{J_* = j\} = \frac{1}{C_\Phi} \int_{\mathcal{M}} h(s) \int_{N_{s\mathcal{M}}} \mathbf{1} \left\{ \varphi_{j,s}(x) > \max_{i \neq j} \varphi_{i,s}(x) \right\} e^{-\Phi_s(x)} dx dV_{\mathcal{M}}(s). \quad (3.16)$$

Remark 3.3. Convergence of the maximum phase alone does not identify the individual excesses or the minimizing index. The almost-everywhere branch convergence in (3.13) supplies this information; neither continuity of the limiting branches nor local uniform convergence is needed. In fact, (3.11) itself needs only the common degree, the common unit threshold, and Theorem 3.1. If two limiting branches tie on a set of positive $\mathbf{Q}_\Phi$ -measure, this minimum limit still holds, but the location requires a higher-order expansion or an explicit set-valued or tie-breaking analysis. For a common degree and arbitrary positive $\mathbf{b}$, apply the corollary to G_j/b_j ; its conclusions then concern the normalized outputs and their minimum.

Corollary 3.4. *Under Theorem 3.1, suppose additionally that*

$$\overline{F}(r) \underset{r \rightarrow \infty}{\sim} K r^a e^{-Lr^\beta}, \quad K, L, \beta > 0, \quad a \in \mathbb{R}.$$

One may take $w(r) = L\beta r^{\beta-1}$, and

$$\mathbb{P}\{E_\tau(\mathbf{b})\} \underset{\tau \rightarrow \infty}{\sim} KC_\Phi(L\beta)^{-\kappa} r_\tau^{a-\beta\kappa} e^{-Lr_\tau^\beta}. \quad (3.17)$$

In particular, let $Z \sim N_k(0, I_k)$ and let G be continuous and positively homogeneous of degree $\gamma > 0$. Write $g = G|_{\mathbb{S}^{k-1}}$. Assume that $M = \max g > 0$, that its complete maximizing set $\mathcal{M}$ is a compact boundaryless m -dimensional C^2 manifold, and that $c = k - 1 - m \geq 1$. Suppose uniformly in normal coordinates that

$$g(\Psi(s, z)) = M - \frac{1}{2} \langle B_s z, z \rangle + o(\|z\|^2), \quad z \rightarrow 0,$$

where B_s is a continuous field of symmetric positive-definite operators. With $r_u = (u/M)^{1/\gamma}$,

$$\mathbb{P}\{G(Z) > u\} \underset{u \rightarrow \infty}{\sim} \frac{(\gamma M)^{c/2}}{(2\pi)^{(m+1)/2}} r_u^{m-1} e^{-r_u^2/2} \int_{\mathcal{M}} \frac{dV_{\mathcal{M}}(s)}{\sqrt{\det B_s}}. \quad (3.18)$$

The determinant is taken on the normal space in an orthonormal frame. This is the leading Gaussian-chaos term in [7, Corollary 7] and [8, Theorem 1.2]; it is included as a normalization check, not as a new scalar result.

3.1 Weighted angular laws and moving thresholds

The preceding theorem freezes the angular density at a finite positive value on the minimizing manifold. This excludes angular laws whose density vanishes or is singular there. It is more economical to state the extension in terms of the rescaled angular measures themselves. The same formulation permits threshold vectors moving on the local normal scales.

We now drop the positive-density assumption. Let $\mathcal{T}$ be a fixed tubular neighborhood of a compact, boundaryless C^2 reference manifold $\mathcal{M}$, with coordinates $u = \Psi(s, z)$. The reference manifold need not be the exact minimizing set for every τ , but all relevant angular mass will be required to remain in this tube. Let r_τ be a reference radius such that $r_\tau \rightarrow \infty$ as $\tau \rightarrow \infty$, and put

$$\lambda_\tau = r_\tau w(r_\tau) \xrightarrow{\tau \rightarrow \infty} \infty. \quad (3.19)$$

For $s \in \mathcal{M}$, let $D_{\lambda,s} : N_s \mathcal{M} \rightarrow N_s \mathcal{M}$ be invertible and continuous in s , and suppose that

$$\sup_{s \in \mathcal{M}} \|D_{\lambda,s}\| \xrightarrow{\lambda \rightarrow \infty} 0. \quad (3.20)$$

For $u = \Psi(s, z) \in \mathcal{T}$, define the rescaled angular coordinate

$$\Xi_\tau(u) = (s, D_{\lambda_\tau, s}^{-1} z) \in N\mathcal{M}. \quad (3.21)$$

Consider the moving event in (2.6), with required radius ρ_τ from (2.5). On $\mathcal{T}$ put

$$\phi_\tau(u) = w(r_\tau)(\rho_\tau(u) - r_\tau). \quad (3.22)$$

We assume that $\phi_\tau \geq -A$ on $\mathcal{T}$ for some fixed $A < \infty$. This allows a moving threshold to lower the reference radius by order $1/w(r_\tau)$. Let $\mathbb{I}_A = [-A, \infty]$ be the one-point compactification, with $e^{-\infty} = 0$.

Let $a_\tau > 0$ be a deterministic angular normalization. It includes both the anisotropic volume scale and any extra power or slowly varying factor caused by the angular law. Define a Radon measure on $\mathbf{NM} \times \mathbb{I}_A$ by

$$\mathfrak{G}_\tau(B) = \frac{1}{a_\tau} \int_{\mathcal{T}} \mathbf{1} \{ (\Xi_\tau(u), \phi_\tau(u)) \in B \} \mu(du). \quad (3.23)$$

Thus $\mathfrak{G}_\tau$ records simultaneously the rescaled angular measure and the local radial phase. This is the point at which moving singular hyperplanes must be controlled; pointwise density convergence alone is not enough.

Fix $p > 0$. The one-sided Gumbel bound recorded above, together with local uniformity on the bounded interval $[-A, 0]$, yields, for all sufficiently large τ ,

$$\frac{\overline{F}(r_\tau + t/w(r_\tau))}{\overline{F}(r_\tau)} \leq C_A(1 + t_+)^{-p}, \quad t \geq -A, \quad (3.24)$$

where $t_+ = \max(t, 0)$. The order p may be chosen arbitrarily large.

Theorem 3.5. *Suppose that (2.9) and (3.20) hold, and fix an order $p > 0$ for the tightness condition below. Assume that*

$$\mathfrak{G}_\tau \xrightarrow[\tau \rightarrow \infty]{v} \mathfrak{G} \quad (3.25)$$

vaguely on $\mathbf{NM} \times \mathbb{I}_A$, where $\mathfrak{G}$ is a Radon measure. Let $\varrho : \mathbf{NM} \rightarrow [0, \infty)$ be a continuous coercive gauge, meaning that every sublevel set $\{\varrho \leq R\}$ is compact, and assume the weighted tightness condition

$$\lim_{R \rightarrow \infty} \limsup_{\tau \rightarrow \infty} \int_{\{\varrho(\xi) > R\} \times \mathbb{I}_A} (1 + t_+)^{-p} \mathfrak{G}_\tau(d\xi, dt) = 0. \quad (3.26)$$

Suppose also that the angular complement is negligible on the new scale:

$$\int_{S \setminus \mathcal{T}} \overline{F}(\rho_\tau(u)) \mu(du) = o(a_\tau \overline{F}(r_\tau)), \quad \tau \rightarrow \infty. \quad (3.27)$$

The vague convergence and weighted tightness already imply finiteness of the following integral, since $e^{-t} \leq C_{A,p}(1 + t_+)^{-p}$ on $\mathbb{I}_A$. Assume its remaining positivity:

$$C_{\mathfrak{G}} := \int_{\mathbf{NM} \times \mathbb{I}_A} e^{-t} \mathfrak{G}(d\xi, dt) > 0. \quad (3.28)$$

Then

$$\mathbb{P}\{E_\tau(\mathbf{t}_\tau)\} \underset{\tau \rightarrow \infty}{\sim} a_\tau \overline{F}(r_\tau) C_{\mathfrak{G}}. \quad (3.29)$$

Conditionally on $E_\tau(\mathbf{t}_\tau)$,

$$(\Xi_\tau(U), \phi_\tau(U)) \xrightarrow[\tau \rightarrow \infty]{d} \mathbf{Q}_{\mathfrak{G}}, \quad \mathbf{Q}_{\mathfrak{G}}(d\xi, dt) = C_{\mathfrak{G}}^{-1} e^{-t} \mathfrak{G}(d\xi, dt), \quad (3.30)$$

where the variables are extended arbitrarily off $\mathcal{T}$. Moreover,

$$w(r_\tau)(R - \rho_\tau(U)) \Big| E_\tau(\mathbf{t}_\tau) \xrightarrow[\tau \rightarrow \infty]{d} \text{Exp}(1), \quad (3.31)$$

independently of the limiting pair in (3.30).

Theorem 3.5 is a radial-tail transfer statement under a supplied graph limit. It does not construct adapted cells for a singular minimizing set. For comparison, the profile extension in [9, Theorem 3.12] assumes an exponentially dominated rescaled profile. General Gumbel tail ratios are controlled here by (3.24) and weighted angular tightness; we do not invoke that theorem as a substitute for these hypotheses.

The complement condition is deliberately explicit. In a fixed normal form it follows from uniform angular separation and a Potter bound whenever the resulting polynomial remainder is $o(a_\tau)$ as $\tau \rightarrow \infty$. For moving thresholds, however, separation must be verified uniformly; it cannot be inherited automatically from the fixed event.

The smooth theorem is the regular-density case of this result: take $A = 0$, $a_\tau = \lambda_\tau^{-\kappa}$, and $\phi_\tau = \lambda_\tau(f - 1)$. Its limiting graph measure is

$$h(s) \, dV_{\mathcal{M}}(s) \, dx \, \delta_{\Phi_s(x)}(dt).$$

The same integrable bound used in its proof implies weighted tightness; angular separation yields the complement estimate. The weighted statement is useful when density freezing is no longer valid.

Appendix A provides density criteria for verifying the graph limit and tightness, including product-power angular weights. Appendix B checks the resulting constants and conditional laws for Dirichlet orthants.

3.2 Two reusable finite-output normal forms

We return to the smooth fixed-threshold setting and notation preceding Theorem 3.1. In particular, $\mathcal{M}$ is a compact boundaryless manifold, $c = n_S - m \geq 1$, μ has a continuous positive density h near $\mathcal{M}$, and (2.9), (2.11), and (2.12) hold. In both forms below, the active index sets are fixed along $\mathcal{M}$, all coefficient fields are continuous in s , all displayed remainders are uniform in s , and local slackness is uniform. These conditions keep the normal form nondegenerate along the entire minimizing manifold.

Proposition 3.6. *Suppose that one active cost, denoted by f_0 , has the Morse–Bott expansion*

$$f_0(\Psi(s, z)) = 1 + \frac{1}{2} \langle H_s z, z \rangle + o(\|z\|^2), \quad z \rightarrow 0, \quad (3.32)$$

where H_s is positive definite. Suppose that the other active costs satisfy

$$f_j(\Psi(s, z)) - f_0(\Psi(s, z)) = L_{j,s}(z) + O(\|z\|^2), \quad j \in \mathcal{W}, \quad z \rightarrow 0, \quad (3.33)$$

for nonzero linear forms $L_{j,s}$, while all remaining costs are uniformly slack. Define the weak cone

$$\mathcal{C}_s = \{x \in N_s \mathcal{M} : L_{j,s}(x) \leq 0 \text{ for every } j \in \mathcal{W}\}. \quad (3.34)$$

Assume that $\mathcal{C}_s$ has positive Gaussian measure. With $D_{\lambda,s} = \lambda^{-1/2} I$,

$$\Phi_s(x) = \begin{cases} \frac{1}{2} \langle H_s x, x \rangle, & x \in \mathcal{C}_s, \\ \infty, & x \notin \mathcal{C}_s, \end{cases} \quad \kappa = \frac{c}{2}. \quad (3.35)$$

The values of Φ_s on the interfaces $\bigcup_{j \in \mathcal{W}} \{L_{j,s} = 0\}$ are immaterial and may be assigned arbitrarily. Consequently, if Z_s is centered Gaussian on $N_s \mathcal{M}$ with covariance H_s^{-1} , then

$$\mathbb{P}\{E_\tau(\mathbf{b})\} \underset{\tau \rightarrow \infty}{\sim} \overline{F}(r_\tau) \lambda_\tau^{-c/2} (2\pi)^{c/2} \int_{\mathcal{M}} \frac{h(s) \mathbb{P}\{Z_s \in \mathcal{C}_s\}}{\sqrt{\det H_s}} \, dV_{\mathcal{M}}(s). \quad (3.36)$$

Thus several weak constraints contribute one correlated Gaussian cone probability, not in general a product of factors $1/2$.

Proposition 3.7. *Suppose instead that there are r genuinely switching directions. Let*

$$N_s \mathcal{M} = \mathbf{V}_s \oplus \mathbf{W}_s, \quad \dim \mathbf{V}_s = r, \quad \dim \mathbf{W}_s = c - r,$$

be a continuous orthogonal splitting. Suppose that a fixed active set $\mathcal{A}$ has the expansions

$$f_j(\Psi(s, y + w)) = 1 + \ell_{j,s}(y) + \frac{1}{2} \langle H_{j,s} w, w \rangle + o(\|y\| + \|w\|^2), \quad j \in \mathcal{A}, \quad (3.37)$$

as $(y, w) \rightarrow (0, 0)$, uniformly for $y \in \mathbf{V}_s$ and $w \in \mathbf{W}_s$, and that all other costs are locally slack. Assume that

$$\Phi_s(x, z) = \max_{j \in \mathcal{A}} \left\{ \ell_{j,s}(x) + \frac{1}{2} \langle H_{j,s} z, z \rangle \right\} \quad (3.38)$$

obeys, uniformly in s , the explicit coercive bound

$$\Phi_s(x, z) \geq c_1(\|x\| + \|z\|^2)$$

for some $c_1 > 0$. The uniform expansion and local slackness then imply, after shrinking the tube if necessary,

$$f(\Psi(s, y + w)) - 1 \geq c_0(\|y\| + \|w\|^2). \quad (3.39)$$

Then the choice

$$D_{\lambda,s}(x + z) = \lambda^{-1}x + \lambda^{-1/2}z$$

yields

$$\mathbb{P}\{E_\tau(\mathbf{b})\} \underset{\tau \rightarrow \infty}{\sim} \overline{F}(r_\tau) \lambda_\tau^{-(c+r)/2} \int_{\mathcal{M}} h(s) \int_{\mathbf{V}_s \times \mathbf{W}_s} e^{-\Phi_s(x,z)} \, dx \, dz \, dV_{\mathcal{M}}(s). \quad (3.40)$$

The power depends on the rank r of the switching directions, not on the number of active output constraints. The master theorem also allows additional weak branches after one separately verifies that they restrict z to a fixed full-dimensional cone; the corresponding phase is then set equal to infinity outside that cone.

Remark 3.8. For a fixed active set, put

$$P_s = \text{conv}\{\nabla_N f_j(s) : f_j(s) = 1\}.$$

Since $f = \max_j f_j$ has a local minimum, $0 \in P_s$. The span of the minimal face of P_s containing zero determines the switching space, while the exterior gradients determine the weak cone. Indeed, zero lies in the relative interior of its minimal face, so the support function of that face is coercive on its span. Projecting the remaining gradients onto the orthogonal complement yields the weak inequalities, and an exposing normal of the face supplies an interior cone direction. The preceding normal forms require their rank and combinatorial type to remain constant along $\mathcal{M}$. If the active set changes, the cone loses interior, the rank changes, or minimizing strata intersect, a separate local analysis is needed. Comparing powers is justified only after uniform integrable normal forms have been established for the individual contributions; borderline matching may introduce logarithms. Thus Theorem 3.1 is a general transfer theorem under a verified normal form, not a claim that every finite maximum has one fixed polyhedral–quadratic description.

4 Explicit two-output normal forms

Assume in this section that $\mathcal{M}$ is a compact, boundaryless, m -dimensional C^2 submanifold of $\mathcal{S}$. Put

$$c = n_{\mathcal{S}} - m \geq 1.$$

Let $\Psi(s, z)$ be a tubular chart, where $s \in \mathcal{M}$ and $z \in N_s \mathcal{M}$, the c -dimensional normal space. We choose the chart so that its volume Jacobian equals one at $z = 0$. Suppose that μ has a

continuous positive density h with respect to Riemannian volume in a neighborhood of $\mathcal{M}$, and write $h(s) = h(\Psi(s, 0))$.

Set

$$f_i = \frac{q_i}{q_*}, \quad f = f_1 \vee f_2 = \frac{q}{q_*}.$$

Assume that f_1, f_2 are C^2 in a fixed tube around $\mathcal{M}$. All coefficient fields below are continuous in s , and all displayed remainders and lower bounds are uniform in $s \in \mathcal{M}$. Determinants are computed in orthonormal frames of the relevant normal spaces.

Theorem 4.1. *Under (2.9), suppose that one of the following three local regimes holds.*

(S) Slack second constraint. *There are positive-definite symmetric operators H_s on $N_s\mathcal{M}$ such that*

$$f_1(\Psi(s, z)) = 1 + \frac{1}{2}\langle H_s z, z \rangle + o(\|z\|^2), \quad z \rightarrow 0, \quad (4.1)$$

and $f_2 \leq 1 - \eta$ near $\mathcal{M}$ for some $\eta > 0$. Then

$$\mathbb{P}\{E_\tau(b)\} \underset{\tau \rightarrow \infty}{\sim} \bar{F}(r_\tau) \lambda_\tau^{-c/2} C_S, \quad (4.2)$$

where

$$C_S = (2\pi)^{c/2} \int_{\mathcal{M}} \frac{h(s)}{\sqrt{\det H_s}} dV_{\mathcal{M}}(s). \quad (4.3)$$

(W) Weakly active second constraint. *Expansion (4.1) holds, and there is a continuous family of nonzero linear forms L_s on $N_s\mathcal{M}$ such that*

$$f_2(\Psi(s, z)) - f_1(\Psi(s, z)) = L_s(z) + O(\|z\|^2), \quad z \rightarrow 0. \quad (4.4)$$

Then

$$\mathbb{P}\{E_\tau(b)\} \underset{\tau \rightarrow \infty}{\sim} \bar{F}(r_\tau) \lambda_\tau^{-c/2} C_W, \quad C_W = \frac{1}{2} C_S. \quad (4.5)$$

(A) Both constraints strongly active. *There is a continuous unit field $e_s \in N_s\mathcal{M}$; put $W_s = e_s^\perp$, so that $N_s\mathcal{M} = \mathbb{R}e_s \oplus W_s$. Writing a normal coordinate as $ye_s + z$, with $z \in W_s$, suppose that*

$$f_1(\Psi(s, ye_s + z)) = 1 + a_s y + \frac{1}{2}\langle H_{1,s} z, z \rangle + o(|y| + \|z\|^2), \quad (4.6)$$

$$f_2(\Psi(s, ye_s + z)) = 1 - \delta_s y + \frac{1}{2}\langle H_{2,s} z, z \rangle + o(|y| + \|z\|^2), \quad (4.7)$$

as $(y, z) \rightarrow (0, 0)$, where $a_s, \delta_s > 0$. At a genuine two-branch strong minimum, the two positive KKT multipliers force the normal gradients to be antiparallel; hence both gradients vanish on W_s and no linear z -term occurs. The displayed expansions record the resulting second-order hypotheses. Define the reduced Hessian

$$\bar{H}_s = \frac{\delta_s H_{1,s} + a_s H_{2,s}}{a_s + \delta_s} \quad (4.8)$$

on W_s , and assume that it is uniformly positive definite. These conditions and the uniform remainders imply, after shrinking the tube, the local coercivity bound

$$f(\Psi(s, ye_s + z)) - 1 \geq c_0(|y| + \|z\|^2) \quad (4.9)$$

for some $c_0 > 0$. The unit field creates no separate orientability restriction: the first expansion implies $\nabla_N f_1(s) = a_s e_s \neq 0$, so the globally labelled branch f_1 itself orients the switching line. Then

$$\mathbb{P}\{E_\tau(b)\} \underset{\tau \rightarrow \infty}{\sim} \bar{F}(r_\tau) \lambda_\tau^{-(c+1)/2} C_A, \quad (4.10)$$

where

$$C_A = (2\pi)^{(c-1)/2} \int_{\mathcal{M}} \frac{h(s)(a_s^{-1} + \delta_s^{-1})}{\sqrt{\det \bar{H}_s}} dV_{\mathcal{M}}(s). \quad (4.11)$$

When $c = 1$, the determinant and $(2\pi)^0$ in (4.11) are interpreted as one.

Remark 4.2. The roles of the two constraints in parts (S) and (W) may of course be reversed. If several disjoint minimizing manifolds have the same radial cost, their contributions of the same order add. If their exponents κ differ, the contribution with the smallest κ dominates, equivalently, it has the largest exponent $-\kappa$ in the factor $\lambda_\tau^{-\kappa}$. A tie at which both normal gradients vanish leads instead to the phase

$$\frac{1}{2} \max\{\langle H_{1,s}z, z \rangle, \langle H_{2,s}z, z \rangle\},$$

and is not included in Theorem 4.1. It is, however, covered by Theorem 3.1 if the resulting maximum of quadratic forms is uniformly coercive.

5 Centered Gaussian processes on a fixed finite grid

Brownian motion is not needed for the radial reduction. On a fixed finite grid, the law of a centered Gaussian process is determined by its covariance matrix, and whitening converts every squared linear output into a quadratic form of one standard Gaussian vector. The following proposition records this reduction, including singular covariance matrices.

Proposition 5.1. *Let $\mathbf{Z}$ be a centered Gaussian vector in $\mathbb{R}^N$ with covariance matrix Γ of rank $r \geq 1$. Let $M_1, \dots, M_d$ be symmetric positive semidefinite $N \times N$ matrices and put*

$$Y_j = \mathbf{Z}^\top M_j \mathbf{Z}, \quad j = 1, \dots, d. \quad (5.1)$$

Choose a full-column-rank factor $A_\Gamma \in \mathbb{R}^{N \times r}$ such that $\Gamma = A_\Gamma A_\Gamma^\top$, and define

$$Q_j = A_\Gamma^\top M_j A_\Gamma, \quad g_j(v) = v^\top Q_j v, \quad v \in \mathbb{S}^{r-1}. \quad (5.2)$$

For $\mathbf{b} \in (0, \infty)^d$, let

$$q(v) = \max_{1 \leq j \leq d} \left(\frac{b_j}{g_j(v)} \right)^{1/2}, \quad (5.3)$$

where a term is infinity when $g_j(v) = 0$. If $Q_j = 0$ for some j , then $Y_j = 0$ almost surely and the positive-threshold joint event is empty. Henceforth assume that every Q_j is nonzero. Then the feasible angular set is nonempty and $0 < q_ = \min_{v \in \mathbb{S}^{r-1}} q(v) < \infty$. For every $u > 0$, with normalized surface measure σ_{r-1} ,*

$$\mathbb{P}\{Y_j > b_j u, \ j = 1, \dots, d\} = \int_{\mathbb{S}^{r-1}} \bar{F}_{\chi_r}(\sqrt{u} q(v)) \sigma_{r-1}(dv), \quad (5.4)$$

where $\bar{F}_{\chi_r}(x) = \mathbb{P}\{\chi_r > x\}$. Consequently, whenever their corresponding angular hypotheses are verified, the normal-form results above apply with

$$\tau = \sqrt{u}, \quad r_\tau = \tau q_*, \quad w(x) = x, \quad \lambda_\tau = r_\tau w(r_\tau) = u q_*^2. \quad (5.5)$$

The construction is intrinsic: if $\Gamma = \tilde{A}_\Gamma \tilde{A}_\Gamma^\top$ is another full-column-rank factor, then $\tilde{A}_\Gamma = A_\Gamma O$ for an orthogonal matrix O , and the forms, minimizing geometry, and angular integral are carried into those above by $v \mapsto Ov$.

For a centered Gaussian process observed at fixed times $t_1, \dots, t_d$, stack all observed coordinates into $\mathbf{Z}$ and use their joint covariance matrix as Γ . A squared norm of a selected block has the form (5.1) with $M_j = P_j^\top P_j$, where P_j selects that block. Thus, for fixed M_j and $\mathbf{b}$, the process dependence enters only through Γ . For scalar Brownian motion the time covariance is $K_{ij} = \min(t_i, t_j)$; for ν independent components, the (i, j) covariance block is $\min(t_i, t_j) I_\nu$. This covariance is special only because it has a simple independent-increment factorization, leading to explicit weak cones.

Corollary 5.2. Let $\nu \geq 1$, and let $X_1, \dots, X_\nu$ be independent copies of a centered Gaussian process, observed on the fixed set $T = \{t_1, \dots, t_d\}$, where $d \geq 2$. Write

$$K_{ij} = \mathbb{E}\{X_1(t_i)X_1(t_j)\}, \quad Y_j := \sum_{\ell=1}^{\nu} X_\ell^2(t_j),$$

assume $K_{jj} > 0$, $1 \leq j \leq d$, and let $r = \text{rank } K$. Choose $B \in \mathbb{R}^{d \times r}$ with $K = BB^\top$, denote its j th row by $b_j^\top$, and put

$$Q_j = I_\nu \otimes (b_j b_j^\top), \quad g_j(v) = v^\top Q_j v, \quad v \in \mathbb{S}^{\nu r - 1}. \quad (5.6)$$

Define

$$\underline{g}(v) := \min_{1 \leq j \leq d} g_j(v), \quad g_* := \max_{v \in \mathbb{S}^{\nu r - 1}} \underline{g}(v), \quad q_* := g_*^{-1/2}. \quad (5.7)$$

Then $g_* > 0$. Suppose that the reduced quadratic forms satisfy the hypotheses of Theorem 3.1, with exponent κ and maximum phase Φ . Put

$$A_u := \{Y_j > u, \ 1 \leq j \leq d\}, \quad M_u := \min_{1 \leq j \leq d} Y_j, \quad J_u := \min_{1 \leq j \leq d} \argmin Y_j.$$

Thus the location of the minimum on the grid is t_{J_u} . Then

$$\mathbb{P}\{A_u\} \underset{u \rightarrow \infty}{\sim} C_\Phi \left(\frac{u}{g_*}\right)^{-\kappa} \bar{F}_{\chi_{\nu r}} \left(\sqrt{\frac{u}{g_*}}\right) \quad (5.8)$$

$$\underset{u \rightarrow \infty}{\sim} \frac{2^{1-\nu r/2} C_\Phi}{\Gamma(\nu r/2)} \left(\frac{u}{g_*}\right)^{\nu r/2 - 1 - \kappa} \exp\left\{-\frac{u}{2g_*}\right\}. \quad (5.9)$$

If E is a unit exponential random variable, then, conditionally on A_u ,

$$M_u - u \xrightarrow[u \rightarrow \infty]{d} 2g_* E. \quad (5.10)$$

If, in addition, the branchwise convergence in Corollary 3.2 holds, with individual phases $\varphi_1, \dots, \varphi_d$, let $(S, X) \sim \mathbf{Q}_\Phi$ and take E independent of (S, X) . Then, conditionally on A_u ,

$$(Y_j - u)_{j=1}^d \xrightarrow[u \rightarrow \infty]{d} 2g_* (E + \Phi_S(X) - \varphi_{j,S}(X))_{j=1}^d \quad (5.11)$$

in $[0, \infty]^d$. If the limiting maximum of the individual phases is unique $\mathbf{Q}_\Phi$ -almost surely, then

$$(M_u - u, J_u) \mid A_u \xrightarrow[u \rightarrow \infty]{d} (2g_* E, J_*), \quad J_* = \argmax_{1 \leq j \leq d} \varphi_{j,S}(X), \quad (5.12)$$

and E is independent of J_* . Hence the limiting minimum excess is exponential with mean $2g_*$. All quantities in this corollary are determined by K and ν ; the factor B only chooses coordinates.

Remark 5.3. If $K = aa^\top$ has rank one and every $a_j \neq 0$, then jointly $Y_j = a_j^2 V$, where $V \sim \chi_\nu^2$. Put $m_0 = \min_j a_j^2$ and $j_0 = \min \argmin_j a_j^2$. Exactly,

$$A_u = \{V > u/m_0\}, \quad M_u = m_0 V, \quad J_u = j_0.$$

Thus $\mathbb{P}(A_u) = \mathbb{P}\{\chi_\nu^2 > u/m_0\}$ and

$$(M_u - u, J_u) \mid A_u \xrightarrow[u \rightarrow \infty]{d} (2m_0 E, j_0), \quad E \sim \text{Exp}(1).$$

The minimizing angular set is the entire sphere, of codimension zero; this elementary case is handled directly rather than by the positive-codimension normal-form theorem.

Corollary 5.4. *Use the definitions of A_u, M_u, J_u , and Y_j from Corollary 5.2, but replace its abstract normal-form and branchwise assumptions by the following covariance conditions. Assume that K is positive definite and, after relabelling the grid if necessary, put*

$$\sigma^2 := K_{11}, \quad \rho_j := \frac{K_{j1}}{\sigma^2}, \quad 2 \leq j \leq d.$$

Thus ρ_j is the Gaussian regression coefficient of $X_1(t_j)$ on $X_1(t_1)$; it need not be a correlation coefficient. Suppose that $\rho_j^2 \geq 1$ for every $j \geq 2$, and define

$$\mathcal{W} := \{j \geq 2 : \rho_j^2 = 1\}, \quad \mathcal{S}_0 := \{j \geq 2 : \rho_j^2 > 1\}.$$

Define the residual covariance matrix

$$K_{ij}^\circ := K_{ij} - \frac{K_{i1}K_{j1}}{\sigma^2}, \quad 2 \leq i, j \leq d, \quad (5.13)$$

which is positive definite in light of the Schur-complement theorem. Let $\mathbf{Z}^\circ = (Z_2^\circ, \dots, Z_d^\circ)$ be centered Gaussian with covariance matrix K° , and put

$$\pi_0 := \mathbb{P}\{\rho_j Z_j^\circ > 0, j \in \mathcal{W}\}, \quad (5.14)$$

with $\pi_0 = 1$ when $\mathcal{W} = \emptyset$. Write $\mathbf{Z}^{\circ,+}$ for $\mathbf{Z}^\circ$ conditioned on the event in (5.14). Positive definiteness implies a strictly positive density on the weak-coordinate subspace, hence $\pi_0 > 0$. Then

$$\mathbb{P}\{A_u\} \underset{u \rightarrow \infty}{\sim} \pi_0 \mathbb{P}\{\sigma^2 \chi_\nu^2 > u\} \quad (5.15)$$

$$\underset{u \rightarrow \infty}{\sim} \frac{\pi_0}{\Gamma(\nu/2)} \left(\frac{u}{2\sigma^2}\right)^{\nu/2-1} \exp\left\{-\frac{u}{2\sigma^2}\right\}. \quad (5.16)$$

Moreover, conditionally on A_u ,

$$\left(Y_1 - u, \left(\frac{Y_j - u}{2\sqrt{u}}\right)_{j \in \mathcal{W}}, \left(\frac{Y_j - u}{u}\right)_{j \in \mathcal{S}_0}\right) \xrightarrow[u \rightarrow \infty]{d} \left(E_0, (\rho_j Z_j^{\circ,+})_{j \in \mathcal{W}}, (\rho_j^2 - 1)_{j \in \mathcal{S}_0}\right), \quad (5.17)$$

where $E_0 \sim \text{Exp}(1/(2\sigma^2))$ is independent of $\mathbf{Z}^{\circ,+}$. Empty subvectors are omitted. In particular,

$$(M_u - u, J_u) \mid A_u \xrightarrow[u \rightarrow \infty]{d} (E_0, 1). \quad (5.18)$$

Remark 5.5. If $K = \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$, then the Y_j are independent scaled chi-square variables. In this case

$$g_* = \left(\sum_{j=1}^d \sigma_j^{-2}\right)^{-1},$$

the conditional excesses converge to independent exponentials with rates $1/(2\sigma_j^2)$, and

$$\mathbb{P}\{J_u = j \mid A_u\} \xrightarrow[u \rightarrow \infty]{} \frac{\sigma_j^{-2}}{\sum_{i=1}^d \sigma_i^{-2}}.$$

Thus $M_u - u$ is asymptotically exponential with rate $\frac{1}{2} \sum_i \sigma_i^{-2}$ and is independent of the limiting location. For equal variances the limiting location is uniform. This example also shows why a deterministic Brownian-type anchor cannot be asserted for an arbitrary covariance matrix.

Remark 5.6. If $\rho_j^2 < 1$ for some j , the selected first coordinate cannot serve as an anchor at energy σ^2 ; indeed, then $g_* < \sigma^2$. A different coordinate may still satisfy the anchor condition after relabelling. If no coordinate does, several constraints enter at the leading energy level and a polyhedral or stratified normal form may be needed. Likewise, singular residual covariance can produce persistent ties or a lower-dimensional cone. Corollary 5.2 still applies once the corresponding normal form has been verified, but Corollary 5.4 deliberately states the clean nondegenerate regime.

Remark 5.7. Corollary 5.4 is a fixed-covariance result and is not uniform as a regression coefficient approaches the weak boundary. To make the transition precise, consider covariance matrices $K^{(u)} \rightarrow K^{(0)} > 0$ as $u \rightarrow \infty$. For each u , let $X_{1,u}, \dots, X_{\nu,u}$ be independent centered Gaussian copies on the grid with covariance matrix $K^{(u)}$, put $Y_{j,u} := \sum_{\ell=1}^{\nu} X_{\ell,u}^2(t_j)$, and write $A_u = \bigcap_j \{Y_{j,u} > u\}$. Define

$$\sigma_u^2 = K_{11}^{(u)}, \quad \rho_{j,u} = \frac{K_{j1}^{(u)}}{\sigma_u^2}, \quad K_{u,ij}^\circ = K_{ij}^{(u)} - \frac{K_{i1}^{(u)} K_{j1}^{(u)}}{\sigma_u^2}, \quad i, j \geq 2.$$

Let K_0° denote the same Schur complement formed from $K^{(0)}$. Let $\mathcal{T}_0$ be a set of coordinates such that

$$\rho_{j,u} \xrightarrow{u \rightarrow \infty} s_j \in \{-1, 1\}, \quad \beta_{j,u} := (\rho_{j,u}^2 - 1)\sqrt{u} \xrightarrow{u \rightarrow \infty} \beta_j \in \mathbb{R},$$

while

$$\liminf_{u \rightarrow \infty} \min_{\substack{j \geq 2 \\ j \notin \mathcal{T}_0}} (\rho_{j,u}^2 - 1) > 0$$

when the complementary set is nonempty. The remaining coordinates are then uniformly slack. The regression decomposition in the proof yields, conditionally on $Y_{1,u} > u$,

$$\left(\frac{Y_{j,u} - u}{2\sqrt{u}} \right)_{j \in \mathcal{T}_0} \xrightarrow{u \rightarrow \infty} \left(\frac{\beta_j}{2} + s_j Z_j^\circ \right)_{j \in \mathcal{T}_0}, \quad (Z_j^\circ)_{j \in \mathcal{T}_0} \sim N(0, (K_0^\circ)_{\mathcal{T}_0 \mathcal{T}_0}).$$

Consequently,

$$\frac{\mathbb{P}\{A_u\}}{\mathbb{P}\{Y_{1,u} > u\}} \xrightarrow{u \rightarrow \infty} \mathbb{P} \left\{ \frac{\beta_j}{2} + s_j Z_j^\circ > 0, \quad j \in \mathcal{T}_0 \right\}.$$

For one transition coordinate this is $\Phi_N(\beta_j / (2\sqrt{K_{0,jj}^\circ}))$, where Φ_N is the standard normal distribution function. Thus the crossover scale is exactly $(\rho_{j,u}^2 - 1)\sqrt{u} = O(1)$ as $u \rightarrow \infty$. In the one-coordinate case $\beta_j = 0$ yields the weak factor $1/2$, while moving beyond this finite window with $\beta_{j,u} \rightarrow +\infty$ as $u \rightarrow \infty$ recovers the slack factor 1. For several transition coordinates, zero shifts yield their correlated Gaussian cone probability, not in general a product of factors $1/2$.

Remark 5.8. The uniform angular law produced by Gaussian whitening coincides with the symmetrized Dirichlet angular law when every $\alpha_i = 1/2$. This angular coincidence should not be confused with the full Dirichlet specialization: Proposition 5.1 uses quadratic outputs and a chi radial variable, whereas Corollary B.3 permits the degree-one orthant outputs and a general Gumbel radial distribution. When some $\alpha_i \neq 1/2$, the product-power factors in (B.1) contain information beyond covariance. The Gaussian orthant case is obtained from that corollary by taking every $\alpha_i = 1/2$ and $R \sim \chi_k$.

5.1 Brownian motion on a fixed grid

The following consequence of the anchor result isolates the weak-cone mechanism. Appendix C presents a separate two-time calculation across all threshold ratios: it exhibits the slack, weak, and strong regimes and checks their constants explicitly.

Let $B = (B_1, \dots, B_\nu)$ be a ν -dimensional standard Brownian motion, with $\nu \geq 1$, and put

$$Y_\nu(t) = \|B(t)\|^2 = \sum_{\ell=1}^{\nu} B_\ell^2(t). \quad (5.19)$$

Fix $d \geq 2$ and times $0 < t_1 < \dots < t_d$, write $\mathbf{T} = (t_1, \dots, t_d)$, and define

$$A_u^{(d)} = \bigcap_{j=1}^d \{Y_\nu(t_j) > u\}. \quad (5.20)$$

If W is a scalar standard Brownian motion, write

$$\pi_d(\mathbf{T}) = \mathbb{P}\{W(t_j - t_1) > 0, j = 2, \dots, d\}. \quad (5.21)$$

This is the positive-orthant probability of a nondegenerate Gaussian vector for fixed distinct times.

Corollary 5.9. *As $u \rightarrow \infty$,*

$$\mathbb{P}\{A_u^{(d)}\} \sim \pi_d(\mathbf{T}) \mathbb{P}\{Y_\nu(t_1) > u\} \quad (5.22)$$

$$\sim \frac{\pi_d(\mathbf{T})}{\Gamma(\nu/2)} \left(\frac{u}{2t_1}\right)^{\nu/2-1} \exp\left\{-\frac{u}{2t_1}\right\}. \quad (5.23)$$

Moreover, conditionally on $A_u^{(d)}$,

$$\left(Y_\nu(t_1) - u, \left(\frac{Y_\nu(t_j) - u}{2\sqrt{u}}\right)_{j=2}^d\right) \xrightarrow[u \rightarrow \infty]{d} \left(E, (W(t_j - t_1))_{j=2}^d \mid W(t_j - t_1) > 0, j = 2, \dots, d\right), \quad (5.24)$$

where $E \sim \text{Exp}(1/(2t_1))$ is independent of the cone-truncated Gaussian vector. In particular,

$$\min_{1 \leq j \leq d} Y_\nu(t_j) - u \mid A_u^{(d)} \xrightarrow[u \rightarrow \infty]{d} E, \quad (5.25)$$

and the minimum is attained at t_1 with conditional probability tending to one.

5.1.1 The explicit three-output constant

For $d = 3$, put

$$\Delta_1 = t_2 - t_1, \quad \Delta_2 = t_3 - t_2.$$

The correlation of $W(\Delta_1)$ and $W(\Delta_1 + \Delta_2)$ equals $\sqrt{\Delta_1/(\Delta_1 + \Delta_2)}$. Hence the bivariate Gaussian orthant formula [12, Section 2.2] yields

$$\pi_3(\mathbf{T}) = \frac{1}{4} + \frac{1}{2\pi} \arcsin \sqrt{\frac{\Delta_1}{\Delta_1 + \Delta_2}} = \frac{1}{4} + \frac{1}{2\pi} \arctan \sqrt{\frac{\Delta_1}{\Delta_2}}. \quad (5.26)$$

For equally spaced times this is $3/8$, rather than $1/4$. Thus the two later weak constraints do not contribute two independent factors $1/2$. The limiting conditional density of the two later coordinates in (5.24) is

$$\frac{\mathbf{1}\{x > 0, y > 0\}}{2\pi\pi_3(\mathbf{T})\sqrt{\Delta_1\Delta_2}} \exp\left\{-\frac{x^2}{2\Delta_1} - \frac{(y-x)^2}{2\Delta_2}\right\}. \quad (5.27)$$

For the original two-component process, $\nu = 2$, formula (5.23) becomes

$$\mathbb{P}\left\{\min_{1 \leq j \leq 3} Y_2(t_j) > u\right\} \underset{u \rightarrow \infty}{\sim} \pi_3(\mathbf{T}) \exp\left\{-\frac{u}{2t_1}\right\}. \quad (5.28)$$

Remark 5.10. Corollary 5.9 is for fixed distinct times. If times coalesce, the covariance of the Gaussian cone becomes singular, so the result is not uniform. Increasing d along a dense grid is also a different problem: positivity at finitely many times approaches a boundary persistence problem. A continuum of output times therefore requires an additional process localization argument.

6 A uniform growing-dimension radial benchmark

The same homogeneous rule can be considered in varying dimension without changing its threshold convention. Let

$$H_k(z) = \sum_{i=1}^k z_i^2, \quad z \in \mathbb{R}^k, \quad Z^{(k)} \sim N_k(0, I_k),$$

and consider $\mathbb{P}\{H_k(Z^{(k)}) > bu\}$. Here the radial law changes with k . Moreover, $H_k = 1$ on the whole sphere, so the minimizing angular set has codimension zero. The following elementary calculation is therefore a uniform radial benchmark, rather than a growing-dimension extension of the angular localization theorems.

Proposition 6.1. *For an integer $k \geq 1$ and $x > 0$, put*

$$a = \frac{k}{2}, \quad p = a - 1, \quad L_k(x) = \frac{x^{a-1} e^{-x}}{\Gamma(a)}.$$

There is the exact representation

$$\mathbb{P}\{H_k(Z^{(k)}) > 2x\} = L_k(x)R(a, x), \quad R(a, x) = \int_0^\infty e^{-t}(1+t/x)^p dt. \quad (6.1)$$

Whenever $x > p_+ := \max(p, 0)$,

$$|R(a, x) - 1| \leq \frac{|p|}{x - p_+}. \quad (6.2)$$

If $k \geq 2$ and $\beta := 1 - p/x > 0$, the sharper bound is

$$\mathbb{P}\{H_k(Z^{(k)}) > 2x\} = \frac{L_k(x)}{\beta}(1 + \varepsilon), \quad -\frac{p}{x^2\beta^2} \leq \varepsilon \leq 0. \quad (6.3)$$

Fix $0 < b_0 \leq b_1 < \infty$, let $k = k(u) \geq 1$ be integer-valued, and take $b \in [b_0, b_1]$. All the following tail equivalents are uniform in b as $u \rightarrow \infty$. If $k = o(u)$, then

$$\mathbb{P}\{H_k(Z^{(k)}) > bu\} = \frac{(bu/2)^{k/2-1} e^{-bu/2}}{\Gamma(k/2)} \left(1 + O\left(\frac{k}{u}\right)\right). \quad (6.4)$$

More generally, if $k/u \rightarrow c \in [0, b_0)$ as $u \rightarrow \infty$ and

$$\beta_{u,b} = 1 - \frac{k-2}{bu},$$

then

$$\mathbb{P}\{H_k(Z^{(k)}) > bu\} \sim \frac{(bu/2)^{k/2-1} e^{-bu/2}}{\Gamma(k/2)\beta_{u,b}} \sim \frac{b}{b-c} \frac{(bu/2)^{k/2-1} e^{-bu/2}}{\Gamma(k/2)}. \quad (6.5)$$

In this latter regime the conditional excess satisfies

$$\sup_{b \in [b_0, b_1]} \sup_{y \geq 0} \left| \mathbb{P}\left\{ \frac{\beta_{u,b}}{2} (H_k(Z^{(k)}) - bu) > y \mid H_k(Z^{(k)}) > bu \right\} - e^{-y} \right| \xrightarrow{u \rightarrow \infty} 0. \quad (6.6)$$

Thus, for each fixed b , the unscaled excess converges to an exponential law with rate $(b-c)/(2b)$. When $k = o(u)$, this rate is $1/2$.

Proof. The chi-square density and the substitution $s = x + t$ yield (6.1). If $p \geq 0$ and $x > p$, then

$$1 \leq R(a, x) \leq \int_0^\infty e^{-(1-p/x)t} dt = \frac{1}{1-p/x},$$

using $\log(1+v) \leq v$. If $-1 < p < 0$, convexity implies $(1+v)^p \geq 1+pv$, whereas $(1+v)^p \leq 1$ for $v \geq 0$. Consequently $1+p/x \leq R(a,x) \leq 1$. These bounds prove (6.2).

For $p \geq 0$, the inequalities $v - v^2/2 \leq \log(1+v) \leq v$ and $e^{-v} \geq 1-v$ imply

$$\frac{1}{\beta} - \frac{p}{x^2\beta^3} \leq \int_0^\infty e^{-\beta t} e^{-pt^2/(2x^2)} dt \leq R(a,x) \leq \frac{1}{\beta}.$$

Multiplication by β proves (6.3); its lower bound remains valid even when negative. Taking $x = bu/2$ in (6.2) proves (6.4). If $k/u \rightarrow c < b_0$ as $u \rightarrow \infty$, then $\inf_b \beta_{u,b}$ is bounded away from zero. For $k \geq 2$, the absolute relative error in the first expression on the right of (6.5) is at most

$$\frac{2(k-2)}{b^2 u^2 \beta_{u,b}^2} = O(u^{-1}), \quad u \rightarrow \infty.$$

The case $k = 1$ follows from (6.2). Finally, $\beta_{u,b} \rightarrow 1 - c/b$ uniformly as $u \rightarrow \infty$, proving both equivalents.

For the excess, fix $y \geq 0$, write $x = bu/2$ and $d = y/\beta_{u,b}$, and divide the tail at $2(x+d)$ by the tail at $2x$. Uniformly in b , the leading factors have ratio

$$e^{-d}(1+d/x)^p = \exp\{-\beta_{u,b}d + O_y((k+1)/u^2)\} \xrightarrow{u \rightarrow \infty} e^{-y}.$$

For $k \geq 2$, (6.3) also implies $R(a, x+d)/R(a, x) \rightarrow 1$ uniformly as $u \rightarrow \infty$: both error bounds vanish and the two corresponding values of $1-p/x$ differ by $pd/[x(x+d)] = O_y((k+1)/u^2)$. For $k = 1$, use (6.2) instead. This proves the convergence uniformly in b for each fixed y . Monotonicity of the survival functions and continuity of e^{-y} , first on a finite mesh and then beyond a sufficiently large mesh endpoint, make the convergence uniform over $y \geq 0$. $\square$

The exact factor $\Gamma(k/2)$ is retained throughout. Replacing it by a Stirling equivalent additionally requires $k \rightarrow \infty$ and introduces a separate relative error of order $1/k$. Neither a uniform positive-codimension angular theorem nor a regime with k/u approaching the threshold b is asserted here.

For a general homogeneous family H_k , extending the angular results requires dimension-uniform control of the rescaled phase and amplitude, as well as a complement estimate relative to the full leading term. In particular, fixed-shift Gumbel convergence alone does not control the growing range of radial shifts selected by increasing normal dimension. These are separate requirements from scalar threshold uniformity: if a fixed random variable has tail equivalent $A(v)$, then the same equivalent at $v = bu$ is already uniform on any interval $0 < b_0 \leq b \leq b_1 < \infty$.

For smooth nondegenerate minima, Katsevich [10, Theorem 2.12 and Corollary 2.17] obtains high-dimensional Laplace expansions with small parameter d^2/λ under quantitative derivative and localization bounds. Alexander and Anya Katsevich [11, Theorem 3.2] expand the logarithm of the normalized integral. Their order- L remainder has scale $(R^2 d)^{L+1}/\lambda^L$, with R subject to localization conditions; when R remains bounded this reduces to d^{L+1}/λ^L . Thus any use of those rates must retain the corresponding regularity and localization assumptions. These smooth results do not directly supply dimension-uniform bounds for switching branches or singular angular weights. The benchmark above is an elementary special case, with an explicit error bound, rather than a claim to extend that high-dimensional theory.

7 Proofs

7.1 Abstract normal-form results

Proof of Theorem 3.1. Let $J(s, z)$ denote the volume Jacobian of the tubular chart and set

$$\begin{aligned} \mathcal{A}_{\lambda,s} &= \{x \in N_s \mathcal{M} : D_{\lambda,s} x \text{ belongs to the chart of } \mathcal{T}\}, \\ A_\tau(s, x) &= \lambda_\tau [f(\Psi(s, D_{\lambda_\tau, s} x)) - 1]. \end{aligned}$$

Every compact subset of the normal bundle is eventually contained in the domains $\mathcal{A}_{\lambda,s}$, uniformly in s . Since $r_\tau f = r_\tau + A_\tau/w(r_\tau)$ and $|\det D_{\lambda_\tau,s}| = \lambda_\tau^{-\kappa}$, the exact local integral is

$$\begin{aligned} & \frac{\mathbb{P}\{E_\tau(\mathbf{b}), U \in \mathcal{T}\}}{\bar{F}(r_\tau)\lambda_\tau^{-\kappa}} \\ &= \int_{\mathcal{M}} \int_{\mathcal{A}_{\lambda_\tau,s}} \frac{\bar{F}(r_\tau + A_\tau(s,x)/w(r_\tau))}{\bar{F}(r_\tau)} h(\Psi(s, D_{\lambda_\tau,s}x)) J(s, D_{\lambda_\tau,s}x) \, dx \, dV_{\mathcal{M}}(s). \end{aligned} \quad (7.1)$$

The Gumbel ratio converges locally uniformly in its shift: this follows from its monotonicity and the continuity of e^{-t} . Compactness of $\mathcal{M}$, continuity of h and J , and uniform contraction of $D_{\lambda,s}$ yield, uniformly on bounded rescaled sets,

$$h(\Psi(s, D_{\lambda_\tau,s}x)) J(s, D_{\lambda_\tau,s}x) \xrightarrow{\tau \rightarrow \infty} h(s).$$

At a point where $\Phi_s(x) = \infty$, we have $A_\tau(s, x) \rightarrow \infty$ as $\tau \rightarrow \infty$, and the Potter bound implies

$$\frac{\bar{F}(r_\tau + A_\tau(s, x)/w(r_\tau))}{\bar{F}(r_\tau)} \leq C_p(1 + A_\tau(s, x))^{-p} \xrightarrow{\tau \rightarrow \infty} 0 = e^{-\Phi_s(x)}.$$

Moreover, (2.12) and (3.4) dominate the integrand in (7.1) by a constant multiple of $(1 + \varrho_s(x))^{-p}$. Equation (3.5) therefore permits dominated convergence on the expanding domains and yields the local contribution C_Φ . On the complement of the tube, (2.11) and (2.12) imply $O(\bar{F}(r_\tau)\lambda_\tau^{-p}) = o(\bar{F}(r_\tau)\lambda_\tau^{-\kappa})$ as $\tau \rightarrow \infty$. This proves (3.7).

For completeness, let H be a bounded continuous function on the normal bundle and let $y \geq 0$. Repeating (7.1) with the angular factor $H(s, x)$ and with the additional event $w(r_\tau)(R - \tau q(U)) > y$ replaces the tail kernel by the same kernel with $A_\tau(s, x)$ replaced by $A_\tau(s, x) + y$. The preceding domination then yields

$$\begin{aligned} & \frac{\mathbb{E} \left[H(S_\tau, D_{\lambda_\tau, S_\tau}^{-1} z_\tau) \mathbf{1}_{E_\tau(\mathbf{b})} \mathbf{1}_{\{U \in \mathcal{T}\}} \mathbf{1}_{\{w(r_\tau)(R - \tau q(U)) > y\}} \right]}{\bar{F}(r_\tau)\lambda_\tau^{-\kappa}} \\ & \xrightarrow{\tau \rightarrow \infty} e^{-y} \int_{\mathcal{M}} h(s) \int_{N_s \mathcal{M}} H(s, x) e^{-\Phi_s(x)} \, dx \, dV_{\mathcal{M}}(s). \end{aligned}$$

The complement estimate above shows that deleting $\mathbf{1}_{\{U \in \mathcal{T}\}}$ changes the numerator by

$$o(\bar{F}(r_\tau)\lambda_\tau^{-\kappa}), \quad \tau \rightarrow \infty.$$

At $y = 0$, the unnormalized angular densities in (7.1) converge in L^1 by the same domination. Normalizing by their total masses therefore proves the stated total-variation convergence; the arbitrary off-tube extension has vanishing conditional mass. In particular, this proves (3.8): for $H \equiv 1$ the limiting mass is C_Φ , so division by (3.7) gives a probability limit, and the preceding convergence for every bounded continuous H is weak convergence. Dividing the general limit by (3.7) proves (3.9) and the asserted independence. $\square$

Proof of Corollary 3.4. The Weibull-tail equivalent implies the Gumbel ratio with $w(r) = L\beta r^{\beta-1}$; substitute $\lambda_\tau = L\beta r_\tau^\beta$ in (3.7). For the Gaussian specialization, $R = \|Z\| \sim \chi_k$ is independent of the uniform angle and

$$\bar{F}_{\chi_k}(r) \underset{r \rightarrow \infty}{\sim} \frac{2^{1-k/2}}{\Gamma(k/2)} r^{k-2} e^{-r^2/2}, \quad h = \frac{\Gamma(k/2)}{2\pi^{k/2}}.$$

Here $q_* = M^{-1/\gamma}$ and $f = (M/g)^{1/\gamma}$ has, as $z \rightarrow 0$, the normal expansion

$$f(\Psi(s, z)) = 1 + \frac{\langle B_s z, z \rangle}{2\gamma M} + o(\|z\|^2).$$

Thus $\kappa = c/2$ and the Gaussian normal integral is $(2\pi\gamma M)^{c/2}/\sqrt{\det B_s}$. Multiplication of the radial and angular constants, using $k - 2 - c = m - 1$, yields (3.18). $\square$

Proof of Corollary 3.2. Work on the fixed tube; its conditional complement is negligible in view of Theorem 3.1. Write

$$X_\tau = D_{\lambda_\tau, S_\tau}^{-1} z_\tau, \quad E_\tau^R = w(r_\tau)(R - \tau q(U)),$$

and, with $U = \Psi(S_\tau, z_\tau)$, put

$$f_{\tau,j} = f_j(U), \quad f_\tau = \max_j f_{\tau,j}.$$

Theorem 3.1 yields the joint conditional convergence

$$(S_\tau, X_\tau, E_\tau^R) \xrightarrow[\tau \rightarrow \infty]{d} (S, X, E), \quad (7.2)$$

where E is unit exponential and independent of (S, X) . On $E_\tau(\mathbf{1}_d)$ all branches are positive, and the minimum has the exact representation

$$M_\tau = \left(\frac{R}{q(U)} \right)^\gamma.$$

Since $q(U) = q_* f_\tau$ and $R = \tau q(U) + E_\tau^R/w(r_\tau)$, the tightness in (7.2), uniform contraction of the normal chart, and continuity of f near $\mathcal{M}$ imply $f_\tau \rightarrow 1$ in conditional probability as $\tau \rightarrow \infty$. Therefore

$$\zeta_\tau(M_\tau - \tau^\gamma) = \frac{\lambda_\tau}{\gamma} \left[\left(1 + \frac{E_\tau^R}{\lambda_\tau f_\tau} \right)^\gamma - 1 \right] \xrightarrow[\tau \rightarrow \infty]{d} E.$$

The same Taylor expansion also yields the sharper conditional relation

$$\zeta_\tau(M_\tau - \tau^\gamma) = E_\tau^R + o_{\mathbb{P}}(1), \quad \tau \rightarrow \infty. \quad (7.3)$$

This proves (3.11) without any branchwise phase assumption.

For the remaining assertions, insert $H((s, x), \varphi_{\lambda_\tau}(s, x))$, with H bounded and continuous, into (7.1), and replace A_τ by $A_\tau + y$ for $y \geq 0$. On $\{\Phi < \infty\}$, (3.13) gives the pointwise limit outside a bundle-volume null set, since $h > 0$; on $\{\Phi = \infty\}$ the radial kernel tends to zero. The existing integrable bound therefore yields the limit

$$e^{-y} \int_{\mathcal{M}} \int_{N_s \mathcal{M}} H((s, x), \varphi(s, x)) h(s) e^{-\Phi_s(x)} \, dx \, dV_{\mathcal{M}}(s).$$

After division by the tail equivalent, this proves

$$(S_\tau, X_\tau, \varphi_{\lambda_\tau}(S_\tau, X_\tau), E_\tau^R) \xrightarrow[\tau \rightarrow \infty]{d} (S, X, \varphi(S, X), E).$$

Since $\mathbf{Q}_\Phi$ has density proportional to $e^{-\Phi}$, $\mathbf{Q}_\Phi\{\Phi < \infty\} = 1$. Thus, at the limiting point, $\max_j \varphi_{j,S}(X) = \Phi_S(X) < \infty$: every component is either finite or $-\infty$, and the case of a $+\infty$ component is null. Since $q(U) = q_* f_\tau$ and $\lambda_\tau = \tau q_* w(r_\tau)$,

$$\frac{R}{\tau q_*} = f_\tau + \frac{E_\tau^R}{\lambda_\tau}. \quad (7.4)$$

The common degree and unit thresholds imply $g_j(U) = (q_* f_{\tau,j})^{-\gamma}$. Hence the following identity is exact on $E_\tau(\mathbf{1}_d)$:

$$\zeta_\tau(G_j(RU) - \tau^\gamma) = \frac{\lambda_\tau}{\gamma} \left[\left(\frac{f_\tau + E_\tau^R/\lambda_\tau}{f_{\tau,j}} \right)^\gamma - 1 \right]. \quad (7.5)$$

For every branch with finite limiting phase, the first-order expansion of the right-hand side, together with (3.12) and (7.2), gives $E + \Phi_S(X) - \varphi_{j,S}(X)$. If $\varphi_{j,S}(X) = -\infty$, then $\lambda_\tau(f_\tau - f_{\tau,j}) \rightarrow$

∞ as $\tau \rightarrow \infty$. Because the compact minimizing manifold lies in the common positivity set, after shrinking the fixed tube if necessary all $f_{\tau,j}$ are bounded above and away from zero. The right-hand side of (7.5) therefore tends to $+\infty$. This proves (3.14).

Under the additional uniqueness assumption, the ordering identity

$$J_\tau = \min_j \operatorname{argmin} G_j(RU) = \min_j \operatorname{argmin} g_j(U) = \min_j \operatorname{argmax} f_j(U) = \min_j \operatorname{argmax} \varphi_{\lambda_\tau, j}(S_\tau, X_\tau)$$

holds on $E_\tau(\mathbf{1}_d) \cap \{U \in \mathcal{T}\}$. The conditional probability of the complementary event tends to zero. At a vector with a unique finite maximum, the tie-broken argmax is continuous in the extended-real topology. The joint branch limit just proved therefore implies the joint conditional convergence of J_τ to J_* . Combining this convergence with (7.3) and (7.2), joint Slutsky yields the asserted minimum–location pair. Its independence follows because E is independent of (S, X) whereas J_* is a function of (S, X) . Integrating its indicator under $\mathbf{Q}_\Phi$ yields (3.16). $\square$

Proof of Theorem 3.5. For $t \in \mathbb{I}_A$, define

$$K_\tau(t) = \frac{\bar{F}(r_\tau + t/w(r_\tau))}{\bar{F}(r_\tau)}, \quad K_\tau(\infty) = 0.$$

Monotonicity of K_τ and continuity of e^{-t} make the Gumbel convergence locally uniform in t . On $[T, \infty]$, monotonicity implies $|K_\tau(t) - e^{-t}| \leq K_\tau(T) + e^{-T}$, whose limiting upper bound is $2e^{-T}$. Letting first $\tau \rightarrow \infty$ on $[-A, T]$ and then $T \rightarrow \infty$ gives

$$\sup_{t \in \mathbb{I}_A} |K_\tau(t) - e^{-t}| \xrightarrow{\tau \rightarrow \infty} 0. \quad (7.6)$$

Put $W(t) = (1 + t_+)^{-p}$, with $W(\infty) = 0$. If $R > 0$ and $0 \leq \psi \leq \mathbf{1}_{\{\varrho > R\}}$ is continuous and compactly supported, vague convergence yields

$$\int \psi(\xi) W(t) \mathfrak{G}(\mathrm{d}\xi, \mathrm{d}t) = \lim_{\tau \rightarrow \infty} \int \psi(\xi) W(t) \mathfrak{G}_\tau(\mathrm{d}\xi, \mathrm{d}t).$$

Taking the supremum over such cutoffs and then using (3.26) shows that the $W\mathfrak{G}$ -mass of $\{\varrho > R\}$ tends to zero as $R \rightarrow \infty$. Its mass on each compact sublevel of ϱ is finite by the Radon property. Consequently $\int W \mathrm{d}\mathfrak{G} < \infty$, and $e^{-t} \leq C_{A,p} W(t)$ proves the finiteness asserted in (3.28). The exact radial identity and (3.23) yield

$$\frac{\mathbb{P}\{E_\tau(\mathbf{t}_\tau), U \in \mathcal{T}\}}{a_\tau \bar{F}(r_\tau)} = \int K_\tau(t) \mathfrak{G}_\tau(\mathrm{d}\xi, \mathrm{d}t). \quad (7.7)$$

Let χ_R be continuous and compactly supported on $\mathbf{NM}$, with $0 \leq \chi_R \leq 1$ and $\chi_R = 1$ on $\{\varrho \leq R\}$. Vague convergence makes $\int \chi_R \mathrm{d}\mathfrak{G}_\tau$ bounded, and (7.6) implies, as $\tau \rightarrow \infty$,

$$\left| \int \chi_R(\xi) (K_\tau(t) - e^{-t}) \mathfrak{G}_\tau(\mathrm{d}\xi, \mathrm{d}t) \right| \leq \sup_{\mathbb{I}_A} |K_\tau - e^{-t}| \int \chi_R \mathrm{d}\mathfrak{G}_\tau = o(1).$$

The function $\chi_R(\xi)e^{-t}$ is continuous with compact support on $\mathbf{NM} \times \mathbb{I}_A$, so (3.25) also gives

$$\int \chi_R(\xi) K_\tau(t) \mathfrak{G}_\tau(\mathrm{d}\xi, \mathrm{d}t) \xrightarrow{\tau \rightarrow \infty} \int \chi_R(\xi) e^{-t} \mathfrak{G}(\mathrm{d}\xi, \mathrm{d}t).$$

The contribution of $\{\varrho > R\}$ tends to zero uniformly in τ by (3.24) and (3.26). Letting $R \rightarrow \infty$ in (7.7), and then using (3.27), proves (3.29).

The same cutoff argument with a bounded continuous test function gives (3.30). Finally, for $y \geq 0$, replacing the radial endpoint $\rho_\tau(U)$ by $\rho_\tau(U) + y/w(r_\tau)$ replaces $K_\tau(t)$ by $K_\tau(t + y)$. Uniformly on $\mathbb{I}_A$,

$$K_\tau(t + y) \xrightarrow{\tau \rightarrow \infty} e^{-(t+y)} = e^{-y} e^{-t},$$

and $K_\tau(t + y) \leq K_\tau(t)$. Dividing the resulting joint limit by (3.29) proves both (3.31) and the asserted independence. $\square$

7.2 Reusable finite-output normal forms

Proof of Proposition 3.6. Use $D_{\lambda,s} = \lambda^{-1/2}I$. Away from the finitely many interfaces $L_{j,s}(x) = 0$, equations (3.32)–(3.33) give the phase in (3.35): it is quadratic on $\mathcal{C}_s$ and infinite outside. The interfaces have Lebesgue measure zero. The determinant contributes $\lambda^{-c/2}$. Since $f \geq f_0$, positive definiteness in (3.32) and the uniform remainder imply, after shrinking the tube, the quadratic coercivity required by Theorem 3.1. Finally,

$$\int_{\mathcal{C}_s} e^{-\langle H_s x, x \rangle / 2} dx = \frac{(2\pi)^{c/2}}{\sqrt{\det H_s}} \mathbb{P}\{Z_s \in \mathcal{C}_s\}.$$

Substitution in (3.7) proves (3.36). $\square$

Proof of Proposition 3.7. Write a normal vector as $y + w \in V_s \oplus W_s$ and use

$$y = \lambda^{-1}x, \quad w = \lambda^{-1/2}z.$$

Equation (3.37) then yields the phase (3.38). The Jacobian is

$$\lambda^{-r} \lambda^{-(c-r)/2} = \lambda^{-(c+r)/2}.$$

The coercive bound following (3.38), together with the uniform expansion, implies (3.39) and hence verifies (3.4). It also makes the phase integral finite. Theorem 3.1 now yields (3.40). $\square$

7.3 The explicit two-output result

Proof of Theorem 4.1. Regimes (S) and (W) are Proposition 3.6 with respectively zero and one weak branch, while regime (A) is Proposition 3.7 with switching rank one. We record the short integral evaluations because they yield the explicit two-output constants. Equivalently, all three cases may be inserted directly into Theorem 3.1. In regime (S), take $D_{\lambda,s} = \lambda^{-1/2}I$. Then $\kappa = c/2$ and $\Phi_s(x) = \langle H_s x, x \rangle / 2$, whose Gaussian integral equals $(2\pi)^{c/2} / \sqrt{\det H_s}$.

The same scaling applies in regime (W). On $\{x : L_s(x) < 0\}$ the limiting phase is $\langle H_s x, x \rangle / 2$, whereas on $\{x : L_s(x) > 0\}$ it is infinity. Thus the limiting domain is a half-space through the origin, with Gaussian mass 1/2.

In regime (A), scale the e_s coordinate by λ^{-1} and the W_s coordinates by $\lambda^{-1/2}$. Then $\kappa = (c+1)/2$ and the phase is

$$\Phi_s(x, v) = \max \left\{ a_s x + \frac{1}{2} \langle H_{1,s} v, v \rangle, -\delta_s x + \frac{1}{2} \langle H_{2,s} v, v \rangle \right\}. \quad (7.8)$$

Put

$$A_i(v) = \frac{1}{2} \langle H_{i,s} v, v \rangle, \quad x_0(v) = \frac{A_2(v) - A_1(v)}{a_s + \delta_s}.$$

With $t = x - x_0(v)$, the phase has the exact leading-form decomposition

$$\Phi_s(x, v) = \frac{1}{2} \langle \overline{H}_s v, v \rangle + \max\{a_s t, -\delta_s t\}.$$

Uniform positivity of $\overline{H}_s$, a_s , and δ_s bounds this below by $c(|t| + \|v\|^2)$. Since $|x_0(v)| \leq C\|v\|^2$, it also bounds $|x| + \|v\|^2$ from below up to a fixed positive factor. The uniform remainders therefore imply (4.9). For fixed v , splitting the x -integral at the intersection of the two affine functions yields

$$\int_{\mathbb{R}} e^{-\Phi_s(x, v)} dx = (a_s^{-1} + \delta_s^{-1}) \exp \left\{ -\frac{1}{2} \langle \overline{H}_s v, v \rangle \right\}. \quad (7.9)$$

The Gaussian integral over W_s yields (4.11). In the slack and weak regimes, $f \geq f_1$ and (4.1) give the required quadratic coercivity. In the strong regime it is precisely the consequence recorded in (4.9). The conclusion now follows from Theorem 3.1. $\square$

7.4 The Gaussian covariance reduction

Proof of Proposition 5.1. Let $\mathbf{N}_r$ be standard Gaussian in $\mathbb{R}^r$. Then $\mathbf{Z} \stackrel{d}{=} A_\Gamma \mathbf{N}_r$. Write $\mathbf{N}_r = RV$, where $R \sim \chi_r$ and V is uniform on $\mathbb{S}^{r-1}$, independently of R . Equations (5.1)–(5.2) give

$$Y_j \stackrel{d}{=} R^2 V^\top Q_j V, \quad j = 1, \dots, d.$$

Conditioning on V proves (5.4).

If $Q_j = 0$, then the preceding representation implies $Y_j = 0$ almost surely. Otherwise, for positive semidefinite Q_j , the zero set of g_j is $\ker Q_j \cap \mathbb{S}^{r-1}$. These are finitely many proper subspheres, whose union cannot cover $\mathbb{S}^{r-1}$; hence a common feasible direction exists. The extended function q is lower semicontinuous on the compact sphere, finite at such a direction, and bounded away from zero because $g_j(v) \leq \|Q_j\|$. Thus $0 < q_* < \infty$.

The chi tail satisfies (2.9) with $w(x) = x$, so $r_\tau w(r_\tau) = u q_*^2$, which proves (5.5). Any chosen normal-form theorem then applies once its angular hypotheses have been verified.

If $\tilde{A}_\Gamma$ is another full-column-rank factor of Γ , the two factors differ by an orthogonal right factor: $\tilde{A}_\Gamma = A_\Gamma O$. Therefore $\tilde{Q}_j = O^\top Q_j O$. Orthogonal invariance of normalized surface measure shows that the substitution $v \mapsto Ov$ leaves (5.4), the minimum, and all normal-form constants unchanged. $\square$

Proof of Corollary 5.2. For $\ell = 1, \dots, \nu$, let $\mathbf{N}_\ell$ be independent standard Gaussian vectors in $\mathbb{R}^r$. The covariance factorization yields

$$X_\ell(t_j) \stackrel{d}{=} b_j^\top \mathbf{N}_\ell, \quad 1 \leq j \leq d.$$

After stacking $\mathbf{N}_1, \dots, \mathbf{N}_\nu$, this is exactly the quadratic-form representation (5.6). Writing the stacked vector as RU gives

$$R \sim \chi_{\nu r}, \quad Y_j = R^2 g_j(U), \quad q(U) = \underline{g}(U)^{-1/2}.$$

Every Q_j is nonzero because $K_{jj} > 0$. The finite union of their kernels cannot cover the sphere, so some direction has all $g_j > 0$; compactness then implies $g_* > 0$.

Proposition 5.1 applies with

$$\tau = \sqrt{u}, \quad r_\tau = \sqrt{u/g_*}, \quad \lambda_\tau = u/g_*.$$

Theorem 3.1 and the standard chi-tail expansion give (5.8)–(5.9). For $\gamma = 2$ the scale in (3.10) is

$$\zeta_\tau = \frac{q_*^2}{2} = \frac{1}{2g_*}.$$

The minimum limit (5.10) follows from (3.11) without branchwise convergence. Under the additional branchwise hypothesis, the vector and location limits (5.11) and (5.12) follow from the remaining parts of Corollary 3.2. Finally, a different factor of K changes the reduced forms only by an orthogonal transformation, as shown in Proposition 5.1; this proves the last assertion. $\square$

Proof of Corollary 5.4. For $j \geq 2$, Gaussian regression yields, simultaneously for $\ell = 1, \dots, \nu$,

$$X_\ell(t_j) = \rho_j X_\ell(t_1) + \varepsilon_{\ell j}, \quad (7.10)$$

where the vectors $(\varepsilon_{\ell 2}, \dots, \varepsilon_{\ell d})$ are independent copies of $N_{d-1}(0, K^\circ)$ and are independent of $(X_1(t_1), \dots, X_\nu(t_1))$.

We first identify the normal form. Use standard Gaussian coordinates in which $X_\ell(t_1) = \sigma N_{\ell 1}$ and the residual vector in (7.10) is a linear image of $d - 1$ further standard coordinates. On the resulting sphere $\mathbb{S}^{\nu d - 1}$, the single constraint at t_1 satisfies

$$g_1(v) \leq \sigma^2.$$

Equality holds precisely on the manifold $\mathcal{M} \cong \mathbb{S}^{\nu-1}$ on which all residual coordinates vanish. At such a point the other outputs equal $\rho_j^2 \sigma^2$, so the anchor condition shows that

$$g_* = \sigma^2, \quad q_* = \sigma^{-1}.$$

In a normal chart (θ, z) around $\mathcal{M}$, with $\theta \in \mathbb{S}^{\nu-1}$ and $z \in \mathbb{R}^{\nu(d-1)}$,

$$f_1 = (1 - \|z\|^2)^{-1/2} = 1 + \frac{1}{2}\|z\|^2 + o(\|z\|^2), \quad z \rightarrow 0. \quad (7.11)$$

If $\mathcal{L}_j z \in \mathbb{R}^\nu$ denotes the residual perturbation at t_j , then for $j \in \mathcal{W}$,

$$f_j - f_1 = -\frac{\rho_j}{\sigma} \langle \theta, \mathcal{L}_j z \rangle + O(\|z\|^2), \quad z \rightarrow 0, \quad (7.12)$$

and the displayed linear form is nonzero because its squared norm is $\rho_j^2 K_{jj}^\circ / \sigma^2 = K_{jj}^\circ / \sigma^2 > 0$. Every $j \in \mathcal{S}_0$ is uniformly slack. The anchor expansion (7.11) also supplies the required coercivity. For a standard Gaussian normal coordinate, the vector

$$(\langle \theta, \mathcal{L}_j z \rangle)_{j=2}^d$$

is centered Gaussian with covariance K° , independently of θ . Thus the favorable weak cone in Proposition 3.6 has probability π_0 . Its codimension is $c = \nu(d-1)$, and the angular constant is

$$C_\Phi = \pi_0 (2\pi)^{c/2} \frac{|\mathbb{S}^{\nu-1}|}{|\mathbb{S}^{\nu d-1}|} = \pi_0 2^{c/2} \frac{\Gamma(\nu d/2)}{\Gamma(\nu/2)}. \quad (7.13)$$

Combining (7.13) with the $\chi_{\nu d}$ tail, $r_\tau = \sqrt{u}/\sigma$, and $\lambda_\tau = u/\sigma^2$ proves (5.15)–(5.16).

It remains to identify the conditional coordinates on their natural scales. Put

$$\mathbf{X}_1 = (X_1(t_1), \dots, X_\nu(t_1))^\top, \quad \Theta_u = \frac{\mathbf{X}_1}{\|\mathbf{X}_1\|},$$

on $\{Y_1 > u\}$, and let

$$Z_{u,j}^\circ = \sum_{\ell=1}^{\nu} \Theta_{u,\ell} \varepsilon_{\ell j}.$$

Conditionally on Θ_u , the vector $(Z_{u,2}^\circ, \dots, Z_{u,d}^\circ)$ has distribution $N_{d-1}(0, K^\circ)$, which does not depend on Θ_u . It is also independent of $E_u := Y_1 - u$. Moreover,

$$Y_j - u = (\rho_j^2 - 1)u + \rho_j^2 E_u + 2\rho_j \sqrt{u + E_u} Z_{u,j}^\circ + \sum_{\ell=1}^{\nu} \varepsilon_{\ell j}^2. \quad (7.14)$$

Conditionally on $Y_1 > u$,

$$E_u \xrightarrow[u \rightarrow \infty]{d} E_0, \quad E_0 \sim \text{Exp}(1/(2\sigma^2)).$$

Equation (7.14) shows that the weak constraints converge to $\{\rho_j Z_j^\circ > 0, j \in \mathcal{W}\}$, while the slack constraints hold with probability tending to one. The boundary of the weak cone has zero probability, so the event indicators converge in the joint limit and their expectations tend to $\pi_0 > 0$. Dividing the joint test-function limit by this probability proves (5.17).

For $j \in \mathcal{W}$,

$$\frac{Y_j - Y_1}{2\sqrt{u}} \xrightarrow[u \rightarrow \infty]{d} \rho_j Z_j^{\circ,+} > 0,$$

and for $j \in \mathcal{S}_0$,

$$\frac{Y_j - Y_1}{u} \xrightarrow[u \rightarrow \infty]{p} \rho_j^2 - 1 > 0.$$

Hence the minimum is attained at the anchor with conditional probability tending to one, and (5.18) follows. $\square$

7.5 The finite-time Brownian result

Proof of Corollary 5.9. Apply Corollary 5.4 with the first time as the anchor. Brownian covariance yields

$$\sigma^2 = t_1, \quad \rho_j = \frac{\min(t_j, t_1)}{t_1} = 1, \quad j \geq 2,$$

and

$$K_{ij}^\circ = \min(t_i, t_j) - t_1 = \min(t_i - t_1, t_j - t_1), \quad i, j \geq 2.$$

Thus every later constraint is weak, the residual Gaussian vector has the law

$$(W(t_2 - t_1), \dots, W(t_d - t_1)),$$

and its positive-cone probability is $\pi_d(\mathbf{T})$. This includes $\nu = 1$: the anchor direction manifold is then the two-point sphere $\mathbb{S}^0$, and the same residual-cone calculation applies. Equations (5.15)–(5.16) give (5.22)–(5.23), while (5.17) gives (5.24). Finally, (5.18) yields (5.25) and the asserted location of the minimum. $\square$

A Verification criteria for weighted angular laws

We use the notation and geometric setup of the weighted formulation in Section 3.1. The following criteria separate local convergence of singular angular densities from control of the full rescaled integral.

Lemma A.1. *Suppose that the pullback of μ to the tube has a locally integrable density ω :*

$$\mu(d\Psi(s, z)) = \omega(s, z) dV_{\mathcal{M}}(s) dz. \quad (\text{A.1})$$

Put

$$g_\tau(s, x) = \frac{|\det D_{\lambda_\tau, s}|}{a_\tau} \omega(s, D_{\lambda_\tau, s}x) \mathbf{1}\{\Psi(s, D_{\lambda_\tau, s}x) \in \mathcal{T}\}. \quad (\text{A.2})$$

Also put

$$\tilde{\phi}_\tau(s, x) := \phi_\tau(\Psi(s, D_{\lambda_\tau, s}x))$$

whenever the displayed point lies in $\mathcal{T}$, and extend it arbitrarily elsewhere, where $g_\tau = 0$. A sufficient condition for (3.25) is that $g_\tau \rightarrow g_0$ in $L^1_{\text{loc}}(\mathbf{NM})$ as $\tau \rightarrow \infty$ and that

$$\tilde{\phi}_\tau(s, x) \xrightarrow{\tau \rightarrow \infty} \Phi(s, x) \quad (\text{A.3})$$

locally in $g_\tau dV_{\mathcal{M}} dx$ -measure, where $\Phi : \mathbf{NM} \rightarrow [-A, \infty]$ is Borel measurable. Explicitly, for every compact K and $\varepsilon > 0$, this means

$$\int_K g_\tau(s, x) \mathbf{1}\left\{d_\infty(\tilde{\phi}_\tau(s, x), \Phi(s, x)) > \varepsilon\right\} dV_{\mathcal{M}}(s) dx \xrightarrow{\tau \rightarrow \infty} 0, \quad (\text{A.4})$$

where one may take $d_\infty(s, t) = |e^{-s} - e^{-t}|$ on the compact space $[-A, \infty]$. Then

$$\mathfrak{G}(ds, dx, dt) = g_0(s, x) dV_{\mathcal{M}}(s) dx \delta_{\Phi(s, x)}(dt). \quad (\text{A.5})$$

If, in addition, there is a continuous coercive gauge ϱ such that

$$\lim_{R \rightarrow \infty} \limsup_{\tau \rightarrow \infty} \int_{\{\varrho(s, x) > R\}} (1 + (\tilde{\phi}_\tau(s, x))_+)^{-p} g_\tau(s, x) dV_{\mathcal{M}}(s) dx = 0, \quad (\text{A.6})$$

then (3.26) holds. Consequently,

$$C_{\mathfrak{G}} = \int_{\mathbf{NM}} g_0(s, x) e^{-\Phi(s, x)} dV_{\mathcal{M}}(s) dx. \quad (\text{A.7})$$

This criterion isolates the real analytic task: local convergence must be supplemented by tightness of the full rescaled integral.

Lemma A.2. Let B be an $r \times k_0$ matrix of full row rank and let $\gamma_1, \dots, \gamma_r \in (-1, 0)$. For every $M < \infty$,

$$\sup_{\mathbf{a} \in \mathbb{R}^r} \int_{[-M, M]^{k_0}} \prod_{i=1}^r |(Bx + \mathbf{a})_i|^{\gamma_i} dx < \infty. \quad (\text{A.8})$$

The bound is locally uniform on compact subsets of the full-row-rank matrices.

Proposition A.3. Suppose that in tubular coordinates

$$\omega(s, z) = \omega_0(s, z) \prod_{\ell=1}^N |\psi_\ell(s, z)|^{\beta_\ell}, \quad \beta_\ell > -1, \quad (\text{A.9})$$

where ω_0 is continuous and positive near the zero section. Assume that $|\det D_{\lambda, s}| = d_\lambda$ is independent of s and that, for positive numbers $c_{\ell, \lambda}$,

$$c_{\ell, \lambda}^{-1} \psi_\ell(s, D_{\lambda, s} x) \xrightarrow{\lambda \rightarrow \infty} L_{\ell, s}(x) \quad (\text{A.10})$$

locally uniformly away from the zero sets of the limiting forms. Assume that the union of these zero sets has $dV_{\mathcal{M}}(s) dx$ -measure zero. Put

$$a_\lambda = d_\lambda \prod_{\ell=1}^N c_{\ell, \lambda}^{\beta_\ell}. \quad (\text{A.11})$$

Define the nonnegative rescaled product

$$H_\lambda(s, x) = \mathbf{1}\{\Psi(s, D_{\lambda, s} x) \in \mathcal{T}\} \prod_{\ell=1}^N \left| c_{\ell, \lambda}^{-1} \psi_\ell(s, D_{\lambda, s} x) \right|^{\beta_\ell}.$$

Assume explicitly that these products are locally uniformly integrable: for some fixed λ_0 and every compact $K \subset \mathcal{NM}$,

$$\lim_{M \rightarrow \infty} \sup_{\lambda \geq \lambda_0} \int_K H_\lambda(s, x) \mathbf{1}\{H_\lambda(s, x) > M\} dV_{\mathcal{M}}(s) dx = 0. \quad (\text{A.12})$$

Then the densities in (A.2) converge locally in L^1 to

$$g_0(s, x) = \omega_0(s, 0) \prod_{\ell=1}^N |L_{\ell, s}(x)|^{\beta_\ell}. \quad (\text{A.13})$$

Hence Theorem 3.5 applies with $a_\tau = a_{\lambda_\tau}$ once all its remaining hypotheses have been verified, including the uniform lower phase bound, phase convergence, weighted tightness, the complement estimate, and positivity of the limiting integral.

A convenient sufficient criterion for local uniform integrability is that all normalized negative-power *prelimit* factors form one affine system $B_{\lambda, s} x + \mathbf{a}_{\lambda, s}$, with $B_{\lambda, s}$ ranging in a compact subset of the full-row-rank locus, while the nonnegative-power factors are locally uniformly bounded. Apply Lemma A.2 with each negative exponent multiplied by a fixed number slightly larger than one. Uniform comparability with such affine factors is enough. Successive applications also work for a triangular block structure: after integrating one block, every remaining singular factor must be independent of that block. This is the structure used in the Dirichlet proof below.

Separate rank conditions on overlapping blocks do not suffice: $|x|^{-3/4}|x|^{-3/4}$ is not locally integrable, although each one-row block has full rank. Nor does full rank of the limiting forms alone exclude prelimit spikes near their zero sets. A useful sufficient global criterion is $\tilde{\phi}_\tau \geq c\varrho - C$, together with polynomial bounds on the rescaled angular mass of unit cubes and on their sublevel counts. More precisely, suppose each cube $\mathcal{Q}$ has mass at most $C(1 + \inf_{\mathcal{Q}} \varrho)^B$ and at most $C(1 + R)^D$ cubes meet $\{\varrho \leq R\}$. Any order $p > B + D$ in (3.24) then implies weighted tightness by summation over dyadic annuli. The Dirichlet application below verifies both bounds explicitly.

A.1 Proofs of the verification criteria

Proof of Lemma A.1. Let f be continuous with compact support on $\mathbf{N}\mathcal{M} \times \mathbb{I}_A$. The change of variables defining g_τ gives

$$\begin{aligned} \int f(\xi, t) \mathfrak{G}_\tau(d\xi, dt) &= \int_{\mathbf{N}\mathcal{M}} f((s, x), \tilde{\phi}_\tau(s, x)) \\ &\quad \times g_\tau(s, x) dV_{\mathcal{M}}(s) dx. \end{aligned}$$

On the compact projection of the support of f , the phase convergence in $g_\tau dV_{\mathcal{M}} dx$ -measure and uniform continuity of f show that replacing the displayed phase by $\Phi(s, x)$ changes the integral by $o(1)$ as $\tau \rightarrow \infty$. The local L^1 convergence $g_\tau \rightarrow g_0$ as $\tau \rightarrow \infty$ then yields

$$\int f d\mathfrak{G}_\tau \xrightarrow{\tau \rightarrow \infty} \int_{\mathbf{N}\mathcal{M}} f((s, x), \Phi(s, x)) g_0(s, x) dV_{\mathcal{M}}(s) dx.$$

This proves the vague convergence and (A.5); substitution in (3.28) gives (A.7). Finally,

$$\begin{aligned} &\int_{\{\varrho > R\} \times \mathbb{I}_A} (1 + t_+)^{-p} \mathfrak{G}_\tau(d\xi, dt) \\ &= \int_{\{\varrho(s, x) > R\}} \left(1 + (\tilde{\phi}_\tau(s, x))_+\right)^{-p} g_\tau(s, x) dV_{\mathcal{M}}(s) dx. \end{aligned}$$

The assumed tightness of the densities makes the last expression uniformly negligible as $R \rightarrow \infty$, proving (3.26). $\square$

Proof of Lemma A.2. Fix a full-row-rank matrix B_0 and choose a $(k_0 - r) \times k_0$ matrix C_0 such that $T_{B_0} = \begin{pmatrix} B_0 \\ C_0 \end{pmatrix}$ is invertible. For B in a sufficiently small neighborhood of B_0 , $T_B = \begin{pmatrix} B \\ C_0 \end{pmatrix}$ remains invertible, with uniformly bounded inverse and Jacobian. Under the change $z = T_B x$, the cubes lie in one fixed bounded box and the first r coordinates are exactly Bx . Their integrations are bounded by products of

$$\sup_{a \in \mathbb{R}} \int_{-M'}^{M'} |z + a|^{\gamma_i} dz,$$

which are finite because $\gamma_i > -1$. The remaining integrations have finite volume. A finite cover of any compact subset of the full-row-rank locus proves the asserted local uniformity in B . $\square$

Proof of Proposition A.3. For the present λ -indexed family, denote the density in (A.2) by g_λ . Substitution of (A.9) into (A.2), followed by (A.11), writes the rescaled density as

$$g_\lambda(s, x) = \omega_0(s, D_{\lambda, s} x) H_\lambda(s, x).$$

On every compact set the tube indicator in H_λ is eventually one by uniform contraction, and $\omega_0(s, D_{\lambda, s} x) \rightarrow \omega_0(s, 0)$ uniformly as $\lambda \rightarrow \infty$. Away from the union of the limiting zero sets, (A.10) gives

$$H_\lambda(s, x) \xrightarrow{\lambda \rightarrow \infty} \prod_{\ell=1}^N |L_{\ell, s}(x)|^{\beta_\ell}.$$

That union has bundle-volume zero. Since (A.12) and local boundedness of ω_0 make g_λ locally uniformly integrable, Vitali's theorem upgrades the almost-everywhere convergence to local L^1 convergence. Formula (A.13) and the conclusion now follow from Lemma A.1. $\square$

B Dirichlet angular specialization

For a symmetrized Dirichlet vector on $\mathbb{S}^{k-1}$, $k \geq 2$, with parameter $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_k) \in (0, \infty)^k$, the density with respect to surface measure is [4]

$$h_{\boldsymbol{\alpha}}(u) = c_{\boldsymbol{\alpha}} \prod_{i=1}^k |u_i|^{\beta_i}, \quad \beta_i = 2\alpha_i - 1, \quad c_{\boldsymbol{\alpha}} = \frac{\Gamma(\sum_i \alpha_i)}{2 \prod_i \Gamma(\alpha_i)}. \quad (\text{B.1})$$

Consider

$$X = A^{\top} R U, \quad C = (A^{\top})^{-1}, \quad \Sigma = A^{\top} A, \quad (\text{B.2})$$

where A is nonsingular. For the $d = k$ coordinate outputs, fix $\mathbf{b} \in (0, \infty)^k$ and let $\tilde{\mathbf{b}}$ be the unique solution of

$$\min_{x \geq \mathbf{b}} x^{\top} \Sigma^{-1} x. \quad (\text{B.3})$$

Strict convexity implies uniqueness. To identify the active set, put $\boldsymbol{\eta} = \Sigma^{-1} \tilde{\mathbf{b}}$. The KKT conditions are

$$\boldsymbol{\eta} \geq \mathbf{0}, \quad \tilde{\mathbf{b}} \geq \mathbf{b}, \quad \eta_i (\tilde{b}_i - b_i) = 0.$$

Thus $I = \{i : \eta_i > 0\}$ is the unique minimal index set characterized by

$$\tilde{\mathbf{b}}_I = \mathbf{b}_I, \quad \tilde{\mathbf{b}}_J = \Sigma_{JJ} \Sigma_{II}^{-1} \mathbf{b}_I \geq \mathbf{b}_J, \quad \Sigma_{II}^{-1} \mathbf{b}_I > \mathbf{0}_I. \quad (\text{B.4})$$

Conversely, these relations satisfy the KKT conditions, which also proves the characterization; see [5] for this standard quadratic-programming reduction. Put $m_I = |I|$, let $J = \{1, \dots, k\} \setminus I$, and define

$$\mu_* = (\mathbf{b}_I^{\top} \Sigma_{II}^{-1} \mathbf{b}_I)^{1/2}, \quad u_* = \frac{C \tilde{\mathbf{b}}}{\mu_*} \in \mathbb{S}^{k-1}. \quad (\text{B.5})$$

For the fixed ray $X > u \mathbf{b}$, the minimum radius cost is $q_* = \mu_*$ and the minimizing angular set is the singleton $\{u_*\}$.

Assume that F satisfies (2.9) with auxiliary function w . Let $u_n \rightarrow \infty$ as $n \rightarrow \infty$, and put $r_n = \mu_* u_n$, $w_n = w(r_n)$, and $\lambda_n = r_n w_n$. The standard Gumbel domination bound yields, for every fixed $A_0, p > 0$ required below, a constant $C_{A_0, p} < \infty$ such that

$$\frac{\overline{F}(r_n + t/w_n)}{\overline{F}(r_n)} \leq C_{A_0, p} (1 + t_+)^{-p}, \quad t \geq -A_0, \quad (\text{B.6})$$

for all sufficiently large n . Moving thresholds on the natural scales are assumed eventually componentwise positive and satisfy

$$w_n(\mathbf{t}_n - u_n \tilde{\mathbf{b}})_I \xrightarrow{n \rightarrow \infty} \boldsymbol{\xi}_I \in \mathbb{R}^{m_I}, \quad (\text{B.7})$$

$$\left(\frac{w_n}{r_n}\right)^{1/2} (\mathbf{t}_n - u_n \tilde{\mathbf{b}})_J \xrightarrow{n \rightarrow \infty} \boldsymbol{\xi}_J \in [-\infty, \infty)^{|J|}. \quad (\text{B.8})$$

The limits themselves force eventual positivity on the active block and on every residual coordinate with finite ξ_j . Thus the explicit positivity assumption has content only for residual coordinates with $\xi_j = -\infty$; the componentwise formulation is retained for the exact radial reduction. Define

$$L = \{i : \beta_i \neq 0, (C \tilde{\mathbf{b}})_i = 0\}, \quad (\text{B.9})$$

$$L_J^0 = \{i \in L : J \neq \emptyset, C_{iJ} \neq \mathbf{0}_J\}, \quad L_I^0 = L \setminus L_J^0. \quad (\text{B.10})$$

Put

$$L_J^- = \{i \in L_J^0 : \beta_i < 0\}, \quad L_I^- = \{i \in L_I^0 : \beta_i < 0\}.$$

Assume that $C_{L_J^-, J}$ has full row rank, with the evident void convention when $L_J^- = \emptyset$. No additional condition is needed on $C_{L_I^-, I}$: its rows vanish on J , so it has full row rank automatically because C is nonsingular. Thus the only imposed rank condition concerns genuinely singular J -scale angular factors. The smooth quadratic-programming chart has $m_I - 1$ linearly scaled directions and $|J|$ quadratically scaled directions. Hence

$$d_\lambda = \lambda^{-(m_I-1)-|J|/2}. \quad (\text{B.11})$$

A zero Dirichlet factor indexed by L_J^0 first appears on the $\lambda^{-1/2}$ scale, whereas one indexed by L_I^0 first appears on the λ^{-1} scale. Proposition A.3 therefore yields

$$a_\lambda = \lambda^{-(m_I-1)-|J|/2-\frac{1}{2}\sum_{i \in L_J^0} \beta_i - \sum_{i \in L_I^0} \beta_i} \quad (\text{B.12})$$

$$= \lambda^{\kappa_D},$$

$$\kappa_D = 1 - m_I - \frac{|J|}{2} + \frac{|L_J^0|}{2} + |L_I^0| - \sum_{i \in L_J^0} \alpha_i - 2 \sum_{i \in L_I^0} \alpha_i. \quad (\text{B.13})$$

This is exactly the Dirichlet exponent. Thus the altered power is an angular-measure effect, not a correction to the Gumbel radial tail. When $L = \emptyset$, $\kappa_D = -((m_I - 1) + |J|/2)$; the sign is opposite to the abstract exponent κ because the present normalization is written as $\lambda_n^{\kappa_D}$ rather than $\lambda_n^{-\kappa}$.

To make the limiting constant explicit, set

$$\tau_{\text{reg}}^* = \prod_{\substack{i \notin L \\ \beta_i \neq 0}} \mu_*^{-\beta_i} |(C\tilde{\mathbf{b}})_i|^{\beta_i}, \quad (\text{B.14})$$

$$H(\mathbf{y}) = \prod_{i \in L_I^0} |C_{iI} \mathbf{y}_I|^{\beta_i} \prod_{i \in L_J^0} |C_{iJ} \mathbf{y}_J|^{\beta_i}, \quad (\text{B.15})$$

$$S(\mathbf{y}) = \frac{\mathbf{b}_I^\top}{\mu_*} \Sigma_{II}^{-1} \mathbf{y}_I + \frac{1}{2} \mathbf{y}_J^\top (\Sigma^{-1})_{JJ} \mathbf{y}_J. \quad (\text{B.16})$$

Lemma B.1. *Put*

$$\hat{\mathbf{b}}_I = \frac{\mathbf{b}_I}{\mu_*}, \quad \ell = \Sigma_{II}^{-1} \hat{\mathbf{b}}_I, \quad Q = (\Sigma^{-1})_{JJ}, \quad \mathcal{E} = \{v_I : \ell^\top v_I = 0\}. \quad (\text{B.17})$$

The local angular chart is

$$U_\lambda(v_I, v_J) = \frac{u_* + C\left(\frac{\lambda^{-1} v_I}{\lambda^{-1/2} v_J}\right)}{\|u_* + C\left(\frac{\lambda^{-1} v_I}{\lambda^{-1/2} v_J}\right)\|}, \quad v_I \in \mathcal{E}. \quad (\text{B.18})$$

Then $\ell^\top \hat{\mathbf{b}}_I = 1$. For $v_I \in \mathcal{E}$ and $v_J > \xi_J$, define

$$\tau_\xi(v_I) = \max_{i \in I} \frac{\xi_i - v_i}{\hat{b}_i}, \quad \Phi_\xi(v_I, v_J) = \tau_\xi(v_I) + \frac{1}{2} v_J^\top Q v_J. \quad (\text{B.19})$$

If $R = r_n + s/w_n$, then, as $n \rightarrow \infty$, locally uniformly in the chart,

$$w_n(A^\top R U_{\lambda_n} - u_n \tilde{\mathbf{b}})_I = v_I + \hat{\mathbf{b}}_I \left(s - \frac{1}{2} v_J^\top Q v_J \right) + o(1), \quad (\text{B.20})$$

$$\left(\frac{w_n}{r_n} \right)^{1/2} (A^\top R U_{\lambda_n} - u_n \tilde{\mathbf{b}})_J = v_J + o(1). \quad (\text{B.21})$$

Consequently, away from the interfaces $\{v_j = \xi_j\}$ with finite ξ_j , the limiting required radial phase equals Φ_ξ on $v_J > \xi_J$ and is infinite when at least one strict reverse inequality holds. The interface values are immaterial and are assigned the value infinity. Let

$$\mathcal{H}(v_I, v_J) = \prod_{i \in L_I^0} |C_{iI} v_I|^{\beta_i} \prod_{i \in L_J^0} |C_{iJ} v_J|^{\beta_i}. \quad (\text{B.22})$$

With $d\sigma_\mathcal{E}$ denoting Euclidean surface measure on $\mathcal{E}$,

$$\begin{aligned} & \frac{1}{|\Sigma|^{1/2} \|\ell\|} \int_{v_J > \xi_J} \int_{\mathcal{E}} \mathcal{H}(v_I, v_J) e^{-\Phi_\xi(v_I, v_J)} d\sigma_\mathcal{E}(v_I) dv_J \\ &= \frac{1}{|\Sigma|^{1/2}} \int_{\mathbf{y}_I > \xi_I, \mathbf{y}_J > \xi_J} H(\mathbf{y}) e^{-S(\mathbf{y})} d\mathbf{y}_I d\mathbf{y}_J. \end{aligned} \quad (\text{B.23})$$

Lemma B.2. For the present application, let

$$\phi_n(u) = w_n(\rho_n(u) - r_n),$$

where ρ_n is the required radius for the threshold $\mathbf{t}_n$. In the chart (B.18), put

$$\varrho_D(v_I, v_J) = 1 + \|v_I\| + \|v_J\|^2.$$

Let g_n be the density, with respect to $d\sigma_\mathcal{E}(v_I) dv_J$, of the rescaled pullback $a_{\lambda_n}^{-1} U_{\lambda_n}^* \mu$, extended by zero outside the chart domain. After choosing a sufficiently small fixed tube, there are constants $c, C > 0$ and $B \geq 0$ such that, uniformly for all large n ,

$$\phi_n(U_{\lambda_n}(v_I, v_J)) \geq c\varrho_D(v_I, v_J) - C, \quad (\text{B.24})$$

$$\int_{\mathcal{Q}} g_n(v_I, v_J) d\sigma_\mathcal{E}(v_I) dv_J \leq C \left(1 + \inf_{\mathcal{Q}} \varrho_D\right)^B \quad (\text{B.25})$$

for every unit cube $\mathcal{Q} \subset \mathcal{E} \times \mathbb{R}^J$.

Corollary B.3. Assume that A is nonsingular, $\mathbf{b} \in (0, \infty)^k$, $F \in \text{GMDA}(w)$, the vectors $\mathbf{t}_n$ are eventually componentwise positive, the threshold limits (B.7)–(B.8) hold, and $C_{L_J^-, J}$ has the full row rank specified above. Then

$$\begin{aligned} \mathbb{P}\{X > \mathbf{t}_n\} & \underset{n \rightarrow \infty}{\sim} \frac{c\alpha \tau_{\text{reg}}^*}{|\Sigma|^{1/2}} \lambda_n^{\kappa_D} \overline{F}(r_n) \\ & \times \int_{\mathbf{y}_I > \xi_I, \mathbf{y}_J > \xi_J} H(\mathbf{y}) e^{-S(\mathbf{y})} d\mathbf{y}_I d\mathbf{y}_J. \end{aligned} \quad (\text{B.26})$$

Here a lower bound equal to $-\infty$ imposes no restriction.

Define the componentwise scale

$$(\mathbf{d}_n)_I = w_n \mathbf{1}_I, \quad (\mathbf{d}_n)_J = \left(\frac{w_n}{r_n}\right)^{1/2} \mathbf{1}_J. \quad (\text{B.27})$$

Then, for $\mathbf{x} \in [0, \infty)^k$ at continuity points,

$$\mathbb{P}\{\mathbf{d}_n(X - \mathbf{t}_n) > \mathbf{x} \mid X > \mathbf{t}_n\} \xrightarrow{n \rightarrow \infty} \frac{\int_{\mathbf{y}_I > \xi_I + \mathbf{x}_I, \mathbf{y}_J > \xi_J + \mathbf{x}_J} H(\mathbf{y}) e^{-S(\mathbf{y})} d\mathbf{y}}{\int_{\mathbf{y}_I > \xi_I, \mathbf{y}_J > \xi_J} H(\mathbf{y}) e^{-S(\mathbf{y})} d\mathbf{y}}. \quad (\text{B.28})$$

Here $-\infty + x = -\infty$; a residual coordinate with such a lower limit diverges on the displayed scale.

The limiting active and residual blocks are independent: both the integrand and its domain factorize between the I - and J -coordinates.

If $L_I^0 = \emptyset$, the limit in (B.28) factorizes as

$$\begin{aligned} & \exp\{-\ell^\top \mathbf{x}_I\} \\ & \times \frac{\int_{\mathbf{y}_J > \boldsymbol{\xi}_J + \mathbf{x}_J} \prod_{i \in L} |C_{iJ} \mathbf{y}_J|^{\beta_i} e^{-\mathbf{y}_J^\top Q \mathbf{y}_J / 2} d\mathbf{y}_J}{\int_{\mathbf{y}_J > \boldsymbol{\xi}_J} \prod_{i \in L} |C_{iJ} \mathbf{y}_J|^{\beta_i} e^{-\mathbf{y}_J^\top Q \mathbf{y}_J / 2} d\mathbf{y}_J}, \end{aligned} \quad (\text{B.29})$$

with the J -terms omitted when $J = \emptyset$. Consequently, the active limiting excesses are independent exponentials with rates $\ell_i = (\Sigma_{II}^{-1} \mathbf{b}_I / \mu_*)_i$, $i \in I$, and are independent of the extended residual limiting vector.

Remark B.4. Take $\gamma_j = 1$, $G_j(x) = (A^\top x)_j$, U with the symmetrized Dirichlet law of parameter $\boldsymbol{\alpha}$, $r_n = \mu_* u_n$, $\lambda_n = r_n w(r_n)$, and identify the moving-threshold limits $\boldsymbol{\xi}$ with the vector denoted by $\mathbf{q}$ in the companion Dirichlet formulation [6, Theorem 3.1 and Corollary 4.1]. Then (B.26), (B.28), and (B.29) reproduce that tail expansion and conditional excess law, including the power κ_D , the product-power constant, and the negative-power rank conditions. Thus the agreement is exact, not only at the logarithmic or exponential rate.

This specialization also identifies the failure boundary. If the negative-power forms lose rank, the measures in (3.23) need not be tight under the pure power (B.12). Coalescing factors can require a normalization such as $\lambda^{-1} \log \lambda$. The weighted theorem can transfer such a normalization only after graph convergence and tightness, the complement estimate, and positivity of the limiting Laplace integral have been verified; it does not manufacture the logarithm from pointwise convergence.

B.1 Proofs of the Dirichlet specialization

Proof of Lemma B.1. The quadratic-programming identities yield

$$\Sigma^{-1} \tilde{\mathbf{b}} = \mu_*(\boldsymbol{\ell}, \mathbf{0}_J)^\top, \quad C^\top u_* = (\boldsymbol{\ell}, \mathbf{0}_J)^\top. \quad (\text{B.30})$$

Together with (B.18), Taylor expansion of the normalizing denominator proves (B.20) and (B.21). The active equations and (B.7) show that their maximum required radial phase converges to $\tau_{\boldsymbol{\xi}}(v_I) + v_J^\top Q v_J / 2$.

For completeness, the residual branches imply the sharp domain restriction in the statement. For $j \in J$, put

$$\hat{b}_j = \frac{\tilde{b}_j}{\mu_*}, \quad \xi_{n,j} = \left(\frac{w_n}{r_n} \right)^{1/2} (t_{n,j} - u_n \tilde{b}_j),$$

let $\rho_{n,j}$ be the radius required by this branch, and set $\phi_{n,j} = w_n(\rho_{n,j} - r_n)$. With

$$D_n = \left\| u_* + C \begin{pmatrix} \lambda_n^{-1} v_I \\ \lambda_n^{-1/2} v_J \end{pmatrix} \right\|,$$

on every fixed compact set, for all sufficiently large n , the exact identity

$$\phi_{n,j}(U_{\lambda_n}(v_I, v_J)) = \lambda_n(D_n - 1) + D_n \frac{\sqrt{\lambda_n}(\xi_{n,j} - v_j)}{\hat{b}_j + \lambda_n^{-1/2} v_j} \quad (\text{B.31})$$

follows from $A^\top C = I$ and $r_n w_n = \lambda_n$. Thus, locally uniformly away from $v_j = \xi_j$, this branch tends to $-\infty$ when $v_j > \xi_j$ and to $+\infty$ when $v_j < \xi_j$; if $\xi_j = -\infty$, it tends to $-\infty$ on every

compact set. No pointwise limit is asserted on a finite interface. Those hyperplanes have bundle-volume zero, and their values were assigned infinity by convention. This proves the stated limiting phase.

Introduce an auxiliary $e > 0$ only through the analytic identity

$$e^{-\Phi} = \int_0^\infty e^{-(\Phi+e)} \, de, \quad (\text{B.32})$$

and set

$$\mathbf{y}_J = v_J, \quad \mathbf{y}_I = v_I + \widehat{\mathbf{b}}_I(\tau_{\boldsymbol{\xi}}(v_I) + e). \quad (\text{B.33})$$

This maps $\mathcal{E} \times \{v_J > \boldsymbol{\xi}_J\} \times (0, \infty)$ bijectively, up to boundary null sets, onto $\{\mathbf{y}_I > \boldsymbol{\xi}_I, \mathbf{y}_J > \boldsymbol{\xi}_J\}$. Indeed, its inverse is

$$c = \boldsymbol{\ell}^\top \mathbf{y}_I, \quad v_I = \mathbf{y}_I - \widehat{\mathbf{b}}_I c, \quad e = \min_{i \in I} \frac{y_i - \xi_i}{\widehat{b}_i}, \quad v_J = \mathbf{y}_J. \quad (\text{B.34})$$

The active Jacobian and phase identities are

$$d\sigma_{\mathcal{E}}(v_I) \, de = \|\boldsymbol{\ell}\| \, d\mathbf{y}_I, \quad \Phi_{\boldsymbol{\xi}}(v_I, v_J) + e = S(\mathbf{y}). \quad (\text{B.35})$$

Moreover, $C_{iI} \widehat{\mathbf{b}}_I = 0$ for $i \in L_I^0$, so $\mathcal{H}(v_I, v_J) = H(\mathbf{y})$.

It remains to record the tangential surface Jacobian. The unit normal to $\mathcal{E} \times \mathbb{R}^J$ is $n = (\boldsymbol{\ell}, \mathbf{0}_J)^\top / \|\boldsymbol{\ell}\|$. For an invertible map C ,

$$J_{k-1}(C|_{n^\perp}) = |\det C| \|C^{-\top} n\|.$$

In view of (B.30), $C^{-\top} n = u_*/\|\boldsymbol{\ell}\|$; hence this Jacobian equals $|\Sigma|^{-1/2}/\|\boldsymbol{\ell}\|$. Inserting (B.32) and applying (B.33) cancels $\|\boldsymbol{\ell}\|$ and proves (B.23). $\square$

Proof of Lemma B.2. Write $\boldsymbol{\xi}_{n,I} = w_n(\mathbf{t}_n - u_n \widetilde{\mathbf{b}})_I$; this sequence is bounded by (B.7). Put

$$z_n = \begin{pmatrix} \lambda_n^{-1} v_I \\ \lambda_n^{-1/2} v_J \end{pmatrix}, \quad D_n = \|u_* + C z_n\|.$$

Since $\boldsymbol{\ell}^\top v_I = 0$, the cross term vanishes in the normalizing denominator and the identity

$$D_n^2 = 1 + z_n^\top \Sigma^{-1} z_n \quad (\text{B.36})$$

is exact. Let $\rho_{n,i}(u)$ denote the radius required by the i th threshold alone. For every active coordinate $i \in I$, it satisfies the further exact identity

$$\frac{\rho_{n,i}(U_{\lambda_n}(v_I, v_J))}{r_n} = D_n \frac{\lambda_n \widehat{b}_i + \xi_{n,i}}{\lambda_n \widehat{b}_i + v_i}. \quad (\text{B.37})$$

Indeed, $A^\top u_* = \widetilde{\mathbf{b}}/\mu_*$, $A^\top C = I$, and $r_n w_n = \lambda_n$.

The weights $p_i = \ell_i \widehat{b}_i$ are positive and sum to one, while

$$\sum_{i \in I} p_i \frac{v_i}{\widehat{b}_i} = \boldsymbol{\ell}^\top v_I = 0.$$

The map $v_I \mapsto \max_{i \in I} (-v_i/\widehat{b}_i)$ is continuous and positively homogeneous on $\mathcal{E}$. If $\mathcal{E} = \{0\}$, the following claim is immediate from boundedness of $\boldsymbol{\xi}_{n,I}$. Otherwise the map is strictly positive on the unit sphere of $\mathcal{E}$, because a nonzero weighted-zero-sum vector has a negative coordinate. Compactness of that sphere and boundedness of $\boldsymbol{\xi}_{n,I}$ therefore imply, uniformly in n ,

$$\max_{i \in I} \frac{\xi_{n,i} - v_i}{\widehat{b}_i} \geq c_1 \|v_I\| - C_1. \quad (\text{B.38})$$

Choose the fixed tube small enough that all denominators in (B.37) are uniformly comparable with λ_n . Positive definiteness in (B.36) then yields

$$\lambda_n(D_n - 1) \geq c_2 \left(\frac{\|v_I\|^2}{\lambda_n} + \|v_J\|^2 \right).$$

If the maximum in (B.38) is nonnegative, denominator comparability preserves it up to a fixed positive factor. If it is negative, that display forces $\|v_I\|$ to be bounded, and the corresponding fraction is bounded below by a constant. Finally, since ϕ_n is the maximum of the branchwise radial phases, (B.37) yields

$$\begin{aligned} \phi_n(U_{\lambda_n}) &\geq \lambda_n(D_n - 1) + D_n \max_{i \in I} \frac{\lambda_n(\xi_{n,i} - v_i)}{\lambda_n \widehat{b}_i + v_i} \\ &\geq c_3(\|v_I\| + \|v_J\|^2) - C_3. \end{aligned}$$

This proves (B.24) throughout the rescaled fixed tube.

The exact rescaled density makes the remaining estimates transparent. Writing $\bar{\alpha} = \sum_i \alpha_i$, the central-projection surface Jacobian is $|\Sigma|^{-1/2} \|\ell\|^{-1} d_{\lambda_n} D_n^{-k}$. Since $k + \sum_i \beta_i = 2\bar{\alpha}$, on the rescaled chart

$$\begin{aligned} g_n(v_I, v_J) &= \frac{c_{\alpha}}{|\Sigma|^{1/2} \|\ell\|} D_n^{-2\bar{\alpha}} \prod_{\substack{i \notin L \\ \beta_i \neq 0}} |u_{*,i} + (Cz_n)_i|^{\beta_i} \\ &\quad \times \prod_{i \in L_I^0} |C_{iI} v_I|^{\beta_i} \prod_{i \in L_J^0} |C_{iJ} v_J + \lambda_n^{-1/2} C_{iI} v_I|^{\beta_i}. \end{aligned} \quad (\text{B.39})$$

It is zero off that chart. Shrink the fixed tube so that the regular coordinate factors are bounded above and away from zero. The factor $D_n^{-2\bar{\alpha}}$ and the regular products are then uniformly bounded. Thus only the explicitly displayed zero-coordinate powers require an integrability argument. The nonnegative powers are bounded on a unit cube by a fixed polynomial in $1 + \inf_Q \varrho_D$; for example, it is enough to take

$$B = \sum_{i \in L} \max(\beta_i, 0).$$

For the negative powers, apply Lemma A.2 first to $C_{L_J^-, J}$, uniformly in the affine shift generated by v_I , and then to the restriction of $C_{L_I^-, I}$ to $\mathcal{E}$. The latter restriction has full row rank: every one of its rows annihilates $\widehat{\mathbf{b}}_I$, whereas $\ell^\top \widehat{\mathbf{b}}_I = 1$, so no nonzero row combination can be normal to $\mathcal{E}$. Successive integration on each unit cube proves (B.25). $\square$

Proof of Corollary B.3. Take $G_j(x) = (A^\top x)_j$ and $\gamma_j = 1$, $j = 1, \dots, k$, in the weighted moving normal form. Its moving event is then $\{X > \mathbf{t}_n\}$.

We first verify explicitly the local uniform integrability required in (A.12). Choose $q_0 = 1 + \varepsilon > 1$ so close to one that

$$q_0 \beta_i > -1 \quad \text{for every } i \in L \text{ with } \beta_i < 0.$$

On a fixed compact set, the nonnegative-power factors and all smooth factors are uniformly bounded. Apply Lemma A.2, with exponents $q_0 \beta_i$, first to the J -block forms and then to the restrictions of the I -block forms to $\mathcal{E}$. The imposed J -block rank and the automatic restricted I -block rank proved in Lemma B.2 yield, by successive integration,

$$\sup_{\lambda \geq \lambda_0} \int_K H_\lambda^{q_0} d\sigma_{\mathcal{E}} dv_J < \infty$$

for every compact $K \subset \mathcal{E} \times \mathbb{R}^J$. Consequently,

$$\sup_{\lambda \geq \lambda_0} \int_K H_\lambda \mathbf{1}\{H_\lambda > M\} \, d\sigma_{\mathcal{E}} \, dv_J \leq M^{1-q_0} \sup_{\lambda \geq \lambda_0} \int_K H_\lambda^{q_0} \xrightarrow{M \rightarrow \infty} 0,$$

which is exactly (A.12).

Proposition A.3 now yields local L^1 convergence of the rescaled densities. Together with (B.31), this also implies the local phase convergence in rescaled-density measure: off arbitrarily small neighborhoods of the finite interface hyperplanes the convergence is locally uniform, while local L^1 convergence makes the g_n -mass of those neighborhoods uniformly small when their width is small. Lemma A.1 therefore yields the vague graph limit. Its Laplace integral is $c_{\alpha} \tau_{\text{reg}}^*$ times the left-hand side of (B.23); the lemma thus identifies the constant displayed in (B.26).

It remains to verify the non-local hypotheses of Theorem 3.5. In (B.6), make the single choice

$$p > \max \left\{ B + (m_I - 1) + \frac{|J|}{2} + 1, -\kappa_D \right\}.$$

In light of (B.24), $(1 + \phi_n^+)^{-p} \leq C \varrho_D^{-p}$. The number of unit cubes with $\inf_Q \varrho_D \leq R$ is $O(R^{m_I-1+|J|/2})$ as $R \rightarrow \infty$. Summing (B.25) therefore proves (3.26). Since $\ell > 0$, $\ell^\top \widehat{\mathbf{b}}_I = 1$, and $v_I \in \mathcal{E}$,

$$\tau_{\xi}(v_I) \geq \sum_{i \in I} \ell_i \widehat{b}_i \frac{\xi_i - v_i}{\widehat{b}_i} = \ell^\top \xi_I.$$

Together with positive definiteness of Q when $J \neq \emptyset$, this supplies a lower bound for the limiting phase. The active-coordinate minimum and (B.24) imply $\phi_n \geq -A$ on a sufficiently small fixed tube. The same coercivity, the graph-integral identity, and the nonempty limiting domain yield $C_{\mathfrak{G}} > 0$; finiteness is automatic from the vague graph limit and the weighted tightness just proved.

Finally, define the fixed active-only angular cost

$$q_{I,i}(\theta) = \begin{cases} b_i / (A^\top \theta)_i, & (A^\top \theta)_i > 0, \\ \infty, & (A^\top \theta)_i \leq 0, \end{cases} \quad q_I(\theta) = \max_{i \in I} q_{I,i}(\theta).$$

The active-only quadratic program has squared value μ_*^2 and unique minimizing angle u_* . Lower semicontinuity and compactness therefore give, after fixing the tube $\mathcal{T}$ and decreasing a constant $\eta > 0$ if necessary,

$$\inf_{\theta \notin \mathcal{T}} q_I(\theta) \geq \mu_*(1 + 2\eta).$$

Moreover, for $i \in I$,

$$\frac{t_{n,i}}{u_n} = b_i + \frac{\mu_* \xi_{n,i}}{\lambda_n} \xrightarrow{n \rightarrow \infty} b_i$$

uniformly over the finite active set. Hence, for all large n , the moving active-only required radius is at least $(1 - \delta)u_n q_I(\theta)$, where $\delta > 0$ may be chosen so that $(1 - \delta)(1 + 2\eta) \geq 1 + \eta$. Since the full required radius dominates its active-only counterpart,

$$\rho_n(\theta) \geq r_n(1 + \eta), \quad \theta \notin \mathcal{T}.$$

The Potter bound and the same choice of p now yield

$$\int_{\mathcal{S} \setminus \mathcal{T}} \overline{F}(\rho_n(u)) \, \mu(du) \leq C \lambda_n^{-p} \overline{F}(r_n) = o(\lambda_n^{\kappa_D} \overline{F}(r_n)),$$

as $n \rightarrow \infty$, because $p > -\kappa_D$. This proves (3.27). Theorem 3.5 now yields (B.26).

For $\mathbf{x} \in [0, \infty)^k$,

$$\mathbb{P}\{\mathbf{d}_n(X - \mathbf{t}_n) > \mathbf{x} \mid X > \mathbf{t}_n\} = \frac{\mathbb{P}\{X > \mathbf{t}_n + \mathbf{x}/\mathbf{d}_n\}}{\mathbb{P}\{X > \mathbf{t}_n\}},$$

where division is componentwise. The shifted threshold satisfies (B.7)–(B.8) with $\boldsymbol{\xi}$ replaced by $\boldsymbol{\xi} + \mathbf{x}$. All other factors in (B.26) cancel, proving (B.28). When $L_I^0 = \emptyset$, integration over $\{\mathbf{y}_I > \boldsymbol{\xi}_I + \mathbf{x}_I\}$ factorizes into one-dimensional exponential integrals. Taking their ratio yields $e^{-\boldsymbol{\ell}^\top \mathbf{x}_I}$ and proves (B.29), including the stated independence. The convention $-\infty + x = -\infty$ also proves the extended residual assertion. $\square$

C A two-time squared Brownian norm

Let B_1, B_2 be independent standard Brownian motions, put $B(r) = (B_1(r), B_2(r))$, and define

$$Y(r) = B_1^2(r) + B_2^2(r), \quad r > 0. \quad (\text{C.1})$$

Fix $0 < s < t$, put $\Delta = t - s$, and consider

$$p_b(u) = \mathbb{P}\{Y(s) > u, Y(t) > bu\}, \quad b > 0. \quad (\text{C.2})$$

C.1 Four-dimensional radial structure

Let $Z = (Z_1, Z_2, Z_3, Z_4)$ be standard Gaussian in $\mathbb{R}^4$. The independent-increment representation yields

$$B(s) = \sqrt{s}(Z_1, Z_2), \quad (\text{C.3})$$

$$B(t) = \sqrt{s}(Z_1, Z_2) + \sqrt{\Delta}(Z_3, Z_4). \quad (\text{C.4})$$

Consequently,

$$G_1(Z) = s(Z_1^2 + Z_2^2), \quad (\text{C.5})$$

$$G_2(Z) = (\sqrt{s}Z_1 + \sqrt{\Delta}Z_3)^2 + (\sqrt{s}Z_2 + \sqrt{\Delta}Z_4)^2 \quad (\text{C.6})$$

are two homogeneous quadratic outputs and $(Y(s), Y(t)) = (G_1(Z), G_2(Z))$.

Writing $Z = RU$, where R^2 is chi-square with four degrees of freedom and U is uniform on $\mathbb{S}^3$, produces the required common radial factor. This is a global four-dimensional radius; it is not the two-dimensional Bessel radius at either time separately.

At equal thresholds, the minimizing angular manifold is

$$\mathcal{M} = \{(v_1, v_2, 0, 0) : v_1^2 + v_2^2 = 1\} \cong \mathbb{S}^1. \quad (\text{C.7})$$

The proof of Theorem C.1 below verifies this weak-cone geometry.

C.2 The complete fixed-ray asymptotics

Theorem C.1. *For every fixed $b > 0$, as $u \rightarrow \infty$,*

$$p_b(u) \sim \begin{cases} \exp\left\{-\frac{u}{2s}\right\}, & 0 < b < 1, \\ \frac{1}{2} \exp\left\{-\frac{u}{2s}\right\}, & b = 1, \\ C_b u^{-1/2} \exp\{-uI_b\}, & 1 < b < (t/s)^2, \\ \frac{1}{2} \exp\left\{-\frac{bu}{2t}\right\}, & b = (t/s)^2, \\ \exp\left\{-\frac{bu}{2t}\right\}, & b > (t/s)^2, \end{cases} \quad (\text{C.8})$$

where, in the strongly active interval,

$$I_b = \frac{1}{2s} + \frac{(\sqrt{b} - 1)^2}{2\Delta} \quad (\text{C.9})$$

and

$$C_b = \frac{\Delta^{3/2} b^{1/4}}{\sqrt{2\pi}(\sqrt{b} - 1)(t - s\sqrt{b})}. \quad (\text{C.10})$$

In particular,

$$\mathbb{P}\{\min(Y(s), Y(t)) > u\} \underset{u \rightarrow \infty}{\sim} \frac{1}{2} \exp\left\{-\frac{u}{2s}\right\}. \quad (\text{C.11})$$

Remark C.2. The abstract constant in Theorem 4.1(A) recovers the independent density calculation in the proof of Theorem C.1 exactly, not only its power of u . Write $\vartheta = \sqrt{b}$ and put

$$\alpha_b = (2sI_b)^{-1/2}, \quad \beta_b = (\vartheta - 1)(2\Delta I_b)^{-1/2}.$$

The strong-regime minimizing circle is $\mathcal{M}_b = \{(\alpha_b v, \beta_b v) : v \in \mathbb{S}^1\}$. In its geodesic normal chart, the coefficients are

$$\begin{aligned} a &= \frac{\sqrt{s}(\vartheta - 1)}{\sqrt{\Delta}}, & \delta &= \frac{t - s\vartheta}{\vartheta\sqrt{s\Delta}}, \\ H_1 &= 1 - a^2, & H_2 &= 1 - \delta^2, & \overline{H} &= 1 - a\delta = \frac{2sI_b}{\vartheta}, \\ a^{-1} + \delta^{-1} &= \frac{2\sqrt{s}\Delta^{3/2}I_b}{(\vartheta - 1)(t - s\vartheta)}. \end{aligned}$$

Here $c = 2$, the angular density on $\mathbb{S}^3$ is $1/(2\pi^2)$, and the minimizing circle has length 2π . Hence

$$C_A = \frac{2}{\sqrt{\pi}} \frac{\Delta^{3/2} b^{1/4} \sqrt{I_b}}{(\sqrt{b} - 1)(t - s\sqrt{b})}, \quad \frac{C_A}{2^{3/2} \sqrt{I_b}} = C_b.$$

Indeed, $\overline{F}_{\chi_4}(\sqrt{2uI_b})(2uI_b)^{-3/2}C_A \underset{u \rightarrow \infty}{\sim} C_b u^{-1/2} e^{-uI_b}$. The individual scalars H_1 or H_2 can be negative in parts of the strong regime only when $t > 2s$, and they are never simultaneously negative. The reduced Hessian $\overline{H}$ is always positive, which is precisely the condition used in Theorem 4.1.

Corollary C.3. *Let*

$$A_u = \{Y(s) > u, Y(t) > u\}, \quad \Delta = t - s,$$

and, on A_u , define $\Theta_u^\perp = (-\Theta_{u,2}, \Theta_{u,1})$ together with

$$\Theta_u = \frac{B(s)}{\|B(s)\|}, \quad N = B(t) - B(s), \quad Z_{\parallel,u} = \frac{\langle \Theta_u, N \rangle}{\sqrt{\Delta}}, \quad Z_{\perp,u} = \frac{\langle \Theta_u^\perp, N \rangle}{\sqrt{\Delta}}.$$

Then

$$(Y(s) - u, Z_{\parallel,u}, Z_{\perp,u}) \Big| A_u \xrightarrow[u \rightarrow \infty]{d} (E, Z_+, Z_0), \quad (\text{C.12})$$

where the three limiting variables are independent,

$$E \sim \text{Exp}(1/(2s)), \quad Z_0 \sim N(0, 1), \quad Z_+ \stackrel{d}{=} Z \mid \{Z > 0\}, \quad Z \sim N(0, 1).$$

Consequently,

$$\left(Y(s) - u, \frac{Y(t) - u}{2\sqrt{u\Delta}} \right) \Big| A_u \xrightarrow[u \rightarrow \infty]{d} (E, Z_+), \quad (\text{C.13})$$

$$\min\{Y(s), Y(t)\} - u \Big| A_u \xrightarrow[u \rightarrow \infty]{d} E, \quad (\text{C.14})$$

with independent coordinates in (C.13). In particular, $Y(s) - u$ and the excess of the minimum are asymptotically exponential with mean $2s$, whereas the later excess is of order $\sqrt{u}$ and has a half-normal limit after scaling.

Remark C.4. The middle expression in (C.8) is not uniform as $b \downarrow 1$ or $b \uparrow (t/s)^2$; its constant diverges at both ends. The crossover occurs when the distance of b from the relevant critical value is of order $u^{-1/2}$. For example, for each fixed $\beta \in \mathbb{R}$, if $b_u = 1 + \beta u^{-1/2}$, then the argument used at the first weak boundary yields

$$\frac{\mathbb{P}\{Y(s) > u, Y(t) > b_u u\}}{\mathbb{P}\{Y(s) > u\}} \xrightarrow{u \rightarrow \infty} \bar{\Phi}\left(\frac{\beta}{2\sqrt{\Delta}}\right).$$

Here $\bar{\Phi}$ is the standard normal survival function. A complete uniform moving- b theorem would replace the full cusp integral by such Gaussian-linear transition integrals. It is not needed for fixed b .

Remark C.5. Formula (C.11) holds for every fixed $t - s > 0$, but it is not uniform as $t \downarrow s$. Let $t_u > s$ satisfy $\Delta_u = t_u - s \rightarrow 0$ as $u \rightarrow \infty$, and put

$$p_{1,t_u}(u) = \mathbb{P}\{Y(s) > u, Y(t_u) > u\}.$$

The first transition occurs on the scale

$$u\Delta_u \asymp 1, \quad u \rightarrow \infty.$$

More precisely, if $u\Delta_u \rightarrow c \in [0, \infty)$ as $u \rightarrow \infty$, conditioning on $Y(s) > u$ and using the Brownian increment yields the limiting constraint $E + 2\sqrt{c}Z > 0$, where $E \sim \text{Exp}(1/(2s))$ and $Z \sim N(0, 1)$ are independent. Consequently,

$$\mathbb{P}\{E + 2\sqrt{c}Z > 0\} = \frac{1}{2} + \int_{-\infty}^0 e^{\sqrt{c}z/s} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz = \frac{1}{2} + \exp\left\{\frac{c}{2s^2}\right\} \bar{\Phi}\left(\frac{\sqrt{c}}{s}\right),$$

and hence

$$\frac{p_{1,t_u}(u)}{\mathbb{P}\{Y(s) > u\}} \xrightarrow{u \rightarrow \infty} \frac{1}{2} + \exp\left\{\frac{c}{2s^2}\right\} \bar{\Phi}\left(\frac{\sqrt{c}}{s}\right). \quad (\text{C.15})$$

Thus the factor tends to one when $u\Delta_u \rightarrow 0$ as $u \rightarrow \infty$. If instead $u\Delta_u \rightarrow \infty$ as $u \rightarrow \infty$, divide the same conditional constraint by $2\sqrt{u\Delta_u}$; only the standard-normal term remains, so the factor tends to $1/2$. This is a triangular weak-activation problem and lies beyond the fixed-geometry theorem.

C.3 Proofs of the two-time Brownian results

Proof of Theorem C.1. The Gaussian energy of fixed vectors $x = B(s)$ and $y = B(t)$ is

$$\mathcal{I}(x, y) = \frac{\|x\|^2}{2s} + \frac{\|y - x\|^2}{2\Delta}. \quad (\text{C.16})$$

For fixed norms the energy is minimized when x and y have the same direction. After writing $\|x\| = \sqrt{u}r$ and $\|y\| = \sqrt{u}\rho$, the variational problem reduces to

$$\min_{r \geq 1, \rho \geq \sqrt{b}} \left\{ \frac{r^2}{2s} + \frac{(\rho - r)^2}{2\Delta} \right\}. \quad (\text{C.17})$$

The objective is strictly convex. Direct differentiation and the KKT conditions yield the unique scalar minimizers

$$(1, 1), \quad (1, 1), \quad (1, \sqrt{b}), \quad (1, t/s), \quad ((s/t)\sqrt{b}, \sqrt{b})$$

in the five regimes $0 < b < 1$, $b = 1$, $1 < b < (t/s)^2$, $b = (t/s)^2$, and $b > (t/s)^2$, respectively. At the middle corner $(1, \sqrt{b})$ the two lower-bound multipliers are

$$\mu_r = \frac{t - s\sqrt{b}}{s\Delta}, \quad \mu_\rho = \frac{\sqrt{b} - 1}{\Delta}; \quad (\text{C.18})$$

both are positive exactly when $1 < b < (t/s)^2$. The minimum has the three phases

$$\begin{cases} 1/(2s), & b \leq 1, \\ I_b, & 1 < b < (t/s)^2, \\ b/(2t), & b \geq (t/s)^2. \end{cases} \quad (\text{C.19})$$

Every minimizing set is a circle, generated by the common direction of x and y . For strict inequalities in the first and third lines one constraint is slack. At $b = 1$ and $b = (t/s)^2$ the other constraint is weakly active. In the middle interval both multipliers are positive; one normal direction is linear and one is quadratic. The powers in (C.8) therefore agree respectively with the slack, weak, and strong parts of Theorem 4.1.

We calculate the strongly active constant from the joint density. Conditioning on $B(s)$ and integrating the Gaussian increment over its polar angle yields the density

$$f_{s,t}(x, z) = \frac{1}{4s\Delta} \exp \left\{ -\frac{x}{2s} - \frac{x+z}{2\Delta} \right\} I_0 \left(\frac{\sqrt{xz}}{\Delta} \right), \quad x, z > 0, \quad (\text{C.20})$$

where I_0 is the modified Bessel function. Its standard large-argument expansion [3, Eq. 10.40.1] is

$$I_0(v) = \frac{e^v}{\sqrt{2\pi v}} (1 + O(v^{-1})), \quad v \rightarrow \infty. \quad (\text{C.21})$$

Therefore the change of variables $x = u\xi$, $z = u\eta$ yields, locally uniformly for $\xi, \eta > 0$,

$$u^2 f_{s,t}(u\xi, u\eta) \underset{u \rightarrow \infty}{\sim} \frac{u^{3/2}}{4s\sqrt{2\pi\Delta}} (\xi\eta)^{-1/4} \exp\{-u\mathcal{J}(\xi, \eta)\}, \quad (\text{C.22})$$

where

$$\mathcal{J}(\xi, \eta) = \frac{\xi}{2s} + \frac{(\sqrt{\eta} - \sqrt{\xi})^2}{2\Delta}. \quad (\text{C.23})$$

At the corner $(1, b)$, both boundary derivatives are positive exactly when $1 < b < (t/s)^2$, and

$$\mathcal{J}_\xi(1, b) = \frac{t - s\sqrt{b}}{2s\Delta}, \quad \mathcal{J}_\eta(1, b) = \frac{\sqrt{b} - 1}{2\Delta\sqrt{b}}. \quad (\text{C.24})$$

On the integration quadrant $\xi \geq 1$, $\eta \geq b$, the argument of I_0 is at least $u\sqrt{b}/\Delta$. The relative error in (C.21) is therefore uniformly $O(u^{-1})$ there. Moreover, in the strongly active interval, $\mathcal{J}$ has the unique constrained minimum $(1, b)$ and is separated from its minimum off every neighborhood of that corner. Since $\mathcal{J}$ is a positive-definite quadratic form in $(\sqrt{\xi}, \sqrt{\eta})$, it also satisfies $\mathcal{J}(\xi, \eta) \geq c(\xi + \eta)$ on the positive quadrant. For the actual amplitude in (C.22), the substitutions $\xi = 1 + x/u$ and $\eta = b + y/u$ yield

$$\int_1^\infty \int_b^\infty (\xi\eta)^{-1/4} e^{-u\mathcal{J}(\xi, \eta)} d\eta d\xi \underset{u \rightarrow \infty}{\sim} \frac{b^{-1/4} e^{-u\mathcal{J}(1, b)}}{u^2 \mathcal{J}_\xi(1, b) \mathcal{J}_\eta(1, b)}.$$

The preceding separation and coercivity justify localization in this calculation. Applying it to (C.22) yields the prefactor

$$\frac{b^{-1/4}}{4s\sqrt{2\pi\Delta} \mathcal{J}_\xi(1, b) \mathcal{J}_\eta(1, b)} = \frac{\Delta^{3/2} b^{1/4}}{\sqrt{2\pi}(\sqrt{b} - 1)(t - s\sqrt{b})},$$

which proves (C.9) and (C.10).

At the lower weak boundary, use the radial representation (C.3)–(C.6) with $\tau = \sqrt{u}$. Near $(v, 0) \in \mathcal{M}$, write

$$U = \left(\sqrt{1 - a^2 - \gamma^2} v, av + \gamma v^\perp \right).$$

If $q_i = G_i(U)^{-1/2}$ and $q_* = s^{-1/2}$, then, as $(a, \gamma) \rightarrow (0, 0)$,

$$\frac{q_1}{q_*} = 1 + \frac{1}{2}(a^2 + \gamma^2) + o(a^2 + \gamma^2), \quad (\text{C.25})$$

$$\frac{q_2}{q_*} = 1 - \sqrt{\frac{\Delta}{s}} a + O(a^2 + \gamma^2). \quad (\text{C.26})$$

Moreover,

$$\frac{q_1}{q_*} = (1 - a^2 - \gamma^2)^{-1/2} \geq 1 + \frac{1}{2}(a^2 + \gamma^2),$$

so the first-time constraint supplies the required coercivity and separation. Thus $b = 1$ is regime (W). Here $c = 2$, the angular density is $1/(2\pi^2)$, the length of $\mathcal{M}$ is 2π , and the normal Hessian is the identity. Hence $C_W = 1$. Since

$$\bar{F}_{\chi_4}(r) = e^{-r^2/2} \left(1 + \frac{r^2}{2} \right), \quad r_\tau = \sqrt{u/s}, \quad \lambda_\tau = u/s,$$

the weak formula implies the second line of (C.8).

At the upper weak boundary $b = (t/s)^2$, use the Brownian-bridge regression

$$B(s) = \frac{s}{t} B(t) + W, \quad W \perp B(t), \quad W \sim N_2 \left(0, \frac{s\Delta}{t} I_2 \right).$$

Conditionally on $Y(t) > bu$, write $Y(t) = bu + E_t$, where E_t is exponential with rate $1/(2t)$. The boundary identity $(s/t)^2 bu = u$ yields

$$Y(s) - u = \left(\frac{s}{t} \right)^2 E_t + 2 \frac{s}{t} \langle B(t), W \rangle + \|W\|^2.$$

After division by $2\sqrt{us\Delta/t}$, the middle term converges to a standard normal random variable and the remaining terms vanish in probability. This proves the factor $1/2$ at the upper boundary.

For $b < 1$, under $Y(s) > u$ we have $Y(t)/u \rightarrow 1$ in probability as $u \rightarrow \infty$, so the later constraint is asymptotically slack. For $b > (t/s)^2$, under $Y(t) > bu$ the Brownian-bridge representation implies $Y(s)/u \rightarrow (s/t)^2 b > 1$ in probability as $u \rightarrow \infty$. Thus the inactive constraint is satisfied with conditional probability tending to one in both strict slack regimes. This completes the proof. $\square$

Proof of Corollary C.3. Conditionally on $Y(s) > u$, the memoryless property gives $Y(s) = u + E_u$, where E_u has the exponential law in the statement. The direction Θ_u is uniform on $\mathbb{S}^1$ and is independent of E_u . In view of isotropy and independence of the increment N , $Z_{\parallel,u}$ and $Z_{\perp,u}$ are independent standard normal variables, independent of E_u and Θ_u . The remaining constraint is exactly

$$E_u + 2\sqrt{(u + E_u)\Delta} Z_{\parallel,u} + \Delta(Z_{\parallel,u}^2 + Z_{\perp,u}^2) > 0.$$

In light of memorylessness and isotropy, these conditional variables may be realized using common independent variables $E, Z_{\parallel}, Z_{\perp}$ whose laws do not depend on u . On this realization the indicator converges almost surely to $\mathbf{1}\{Z_{\parallel} > 0\}$. Explicitly, for every bounded continuous H ,

$$\begin{aligned} \mathbb{E} [H(E, Z_{\parallel}, Z_{\perp}) \mathbf{1}_{\{A_u\}} \mid Y(s) > u] &\xrightarrow{u \rightarrow \infty} \mathbb{E} [H(E, Z_{\parallel}, Z_{\perp}) \mathbf{1}_{\{Z_{\parallel} > 0\}}], \\ \mathbb{P}\{A_u \mid Y(s) > u\} &\xrightarrow{u \rightarrow \infty} \mathbb{P}\{Z_{\parallel} > 0\} = \frac{1}{2}. \end{aligned}$$

Dividing the first limit by the second is the conditional-ratio form of dominated convergence and yields (C.12). Since $Z_+ > 0$ almost surely, $Y(t) - u \rightarrow \infty$ in conditional probability as $u \rightarrow \infty$, while $Y(s) - u$ is tight. This proves (C.13) and (C.14). $\square$

References

- [1] R. A. Davis and S. I. Resnick. Extremes of moving averages of random variables from the domain of attraction of the double exponential distribution. *Stochastic Processes and their Applications*, 30:41–68, 1988.
- [2] L. de Haan and A. Ferreira. *Extreme Value Theory: An Introduction*. Springer, New York, 2006.
- [3] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, editors. *NIST Digital Library of Mathematical Functions*, <https://dlmf.nist.gov/10.40>.
- [4] K.-T. Fang and B.-Q. Fang. Generalised symmetrised Dirichlet distributions. In K.-T. Fang and T. W. Anderson, editors, *Statistical Inference in Elliptically Contoured and Related Distributions*, pages 127–136. Allerton Press, New York, 1990.
- [5] E. Hashorva. Asymptotics and bounds for multivariate Gaussian tails. *Journal of Theoretical Probability*, 18:79–97, 2005.
- [6] E. Hashorva. Exact tail asymptotics of Dirichlet distributions. arXiv:0904.0144v3, 2026, <https://arxiv.org/abs/0904.0144v3>.
- [7] D. A. Korshunov, V. I. Piterbarg, and E. Hashorva. On the asymptotic Laplace method and its application to random chaos. *Mathematical Notes*, 97:878–891, 2015. <https://doi.org/10.1134/S0001434615050235>.
- [8] E. Hashorva, D. Korshunov, and V. I. Piterbarg. Asymptotic expansion of Gaussian chaos via probabilistic approach. *Extremes*, 18:315–347, 2015. <https://doi.org/10.1007/s10687-015-0215-3>.
- [9] P. Ievlev. Laplace Asymptotics near Stratified Minimum Sets. arXiv:2608.02626v1, 2026. <https://arxiv.org/abs/2608.02626v1>.
- [10] Anya Katsevich. The Laplace asymptotic expansion in high dimensions. arXiv:2406.12706v3, revised 12 June 2025 (first posted in 2024). <https://arxiv.org/abs/2406.12706v3>.
- [11] Alexander Katsevich and Anya Katsevich. High-dimensional Laplace asymptotics up to the concentration threshold. arXiv:2602.23151v2, revised 12 March 2026. <https://arxiv.org/abs/2602.23151v2>.
- [12] Y. L. Tong. *The Multivariate Normal Distribution*. Springer, New York, 1990.